



VYSOKÉ UČENÍ FAKULTA  
TECHNICKÉ INFORMAČNÍCH  
V BRNĚ TECHNOLOGIÍ



# 4

## Síťová vrstva

IPK 2021/2022 L

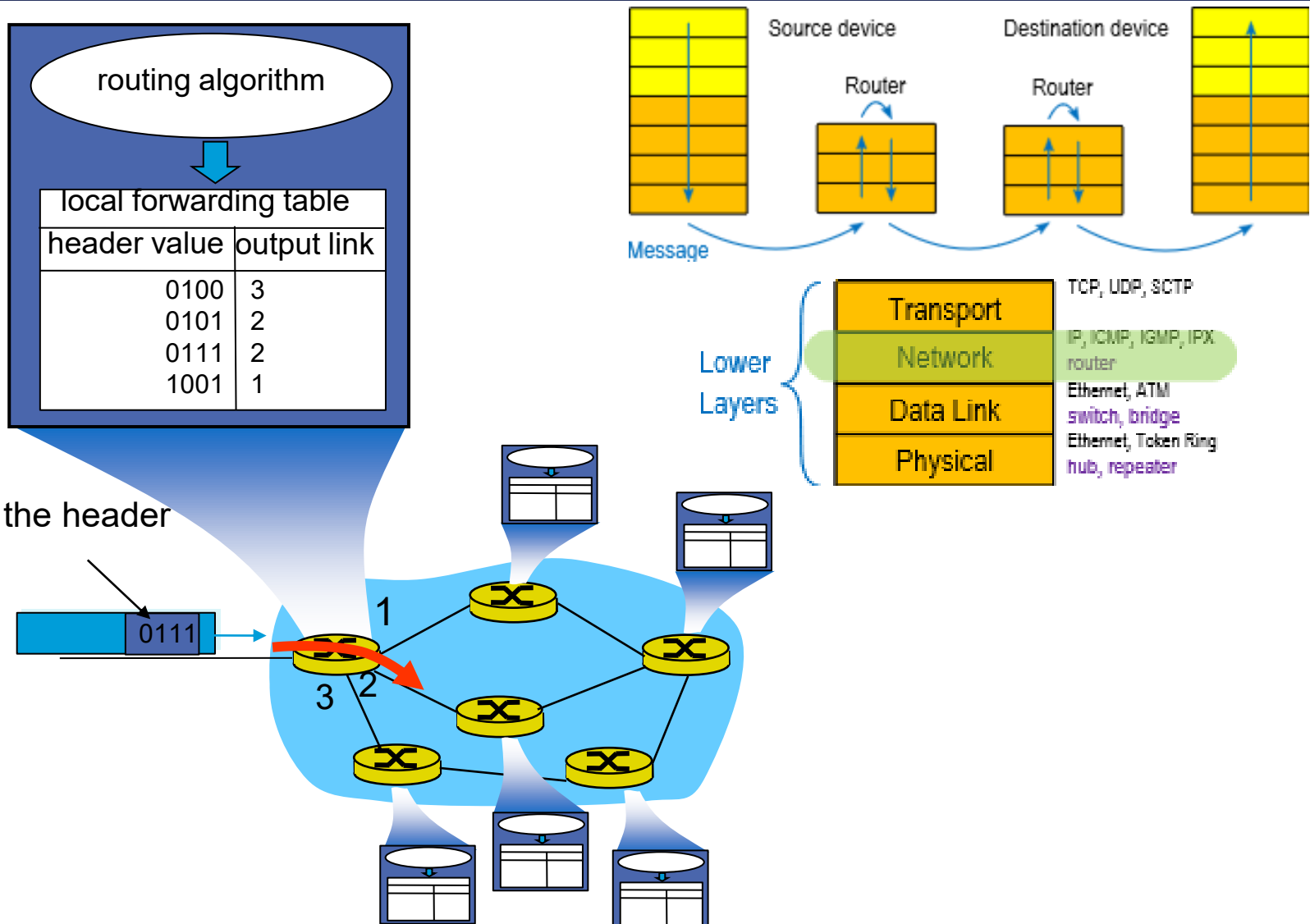
# Obsah

- Služby a funkce síťové vrstvy
- Protokol IPv4
  - Funkce, formát
  - Fragmentace a znovusestavení
  - Adresování v IPv4
- Správa adresního prostoru
- Protokol ICMP
  - Funkce, formát
  - Příklady služeb využívající ICMP
- Protokol DHCP
  - Funkce, formát
- Překlad adres, NAT

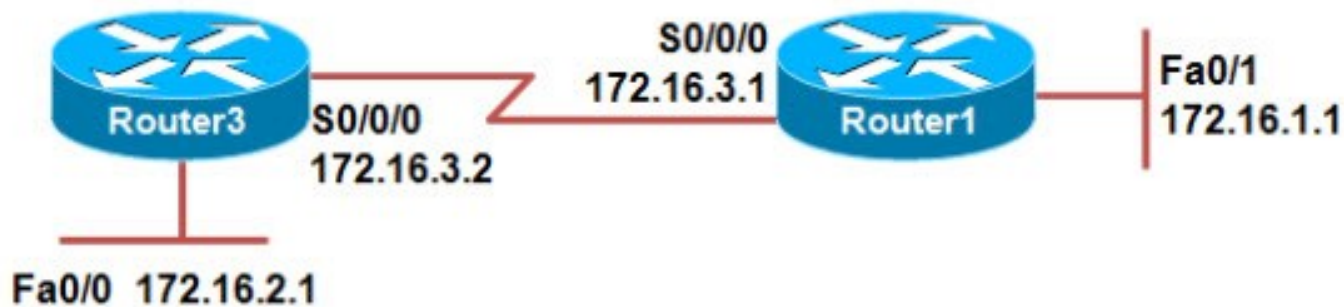
# Klíčové funkce síťové vrstvy

- Přeposílání paketů (**packet forwarding**)
  - Přesunutí (přepsání) paketu ze vstupního rozhraní směrovače na výstupní rozhraní
- Směrování (**routing**)
  - Zjištění cesty (route) pro každý paket, který je přeposílán směrovačem
- Klíčovým prvkem je směrovač (**router**)
  - Pracuje na síťové vrstvě (je adresovatelný IP adresou)
  - Využívá směrovací tabulku (ta určuje kam bude paket přeposlán)

# Směrování



# Reálná směrovací tabulka



```
R3# show ip route
<output omitted>
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

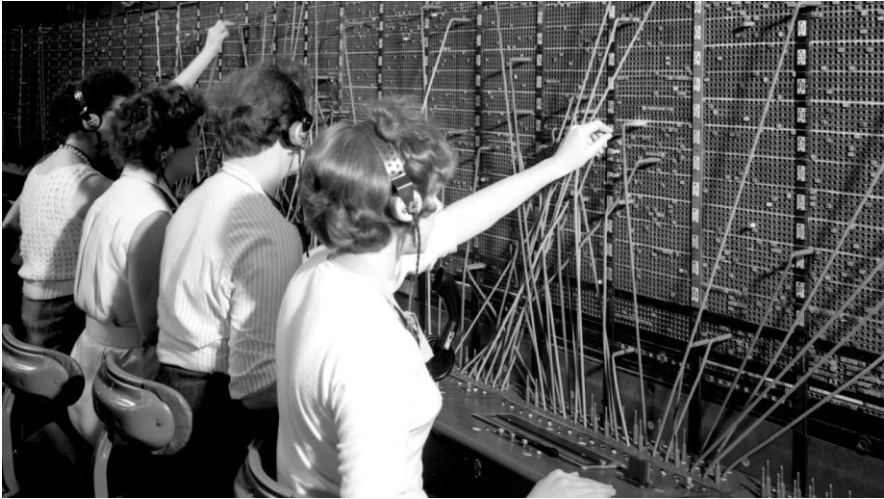
 172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
D    172.16.0.0/16 is a summary, 00:20:29, Null0
D    172.16.1.0/24 [90/2297856] via 172.16.3.1, 00:13:56, Serial0/0/0
C    172.16.2.0/24 is directly connected, FastEthernet0/0
C    172.16.3.0/30 is directly connected, Serial0/0/0
S*   0.0.0.0/0 is directly connected, Serial0/0/0
```

# (Ne)Spojované L3 služby

- Datagramové sítě poskytují nespojovanou službu
  - Best effort delivery
  - Bezstavové směrovače
- Sítě s virtuálními okruhy poskytují spojově orientovanou službu
  - Stavové směrovače
  - Služba: host-host
  - Různé služby
    - Garantované přenosové pásmo
    - Ztráta paketů
    - Pořadí paketů
    - Zpoždění paketů

# CO vs. CL (1)

- Connection-Oriented



- Connection-Less

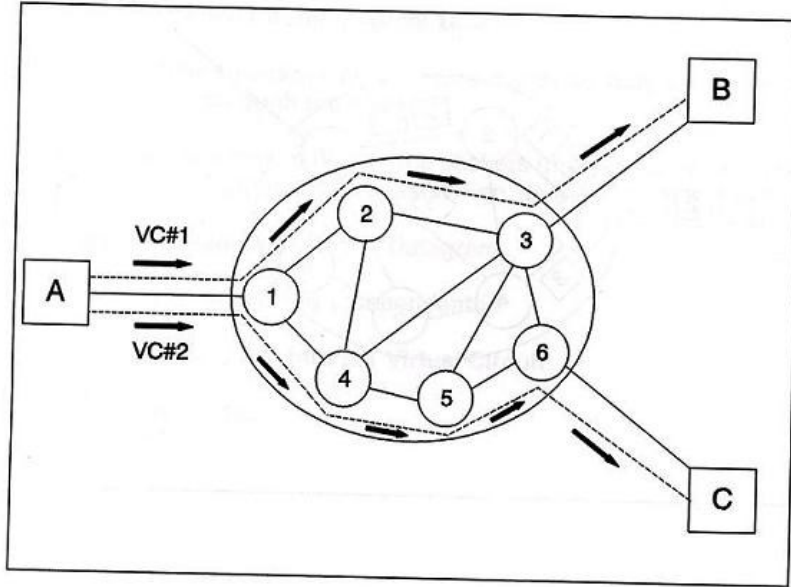


In a few years, men will be able to communicate more effectively through a machine than face to face. That is a rather startling thing to say, but it is our conclusion.

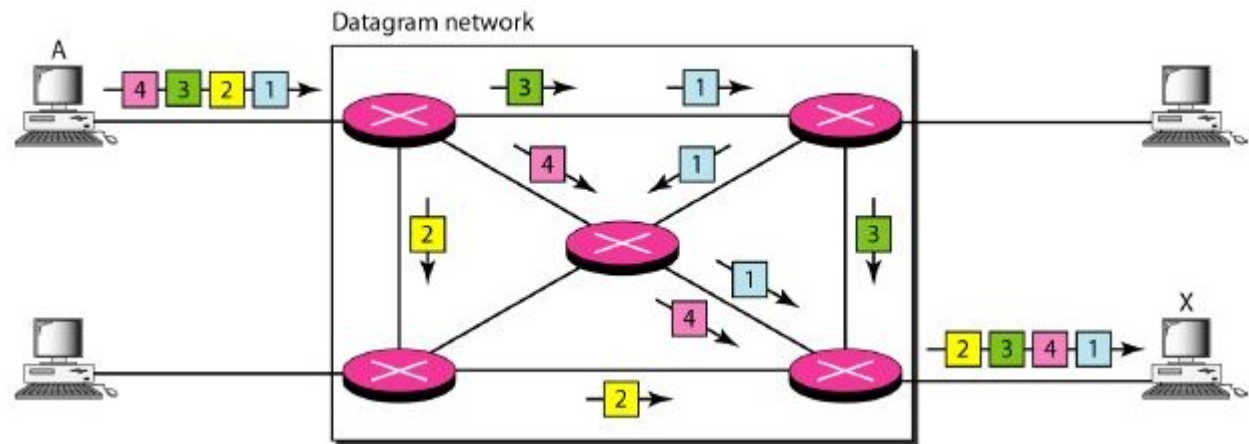
— J. C. R. Licklider —

AZ QUOTES

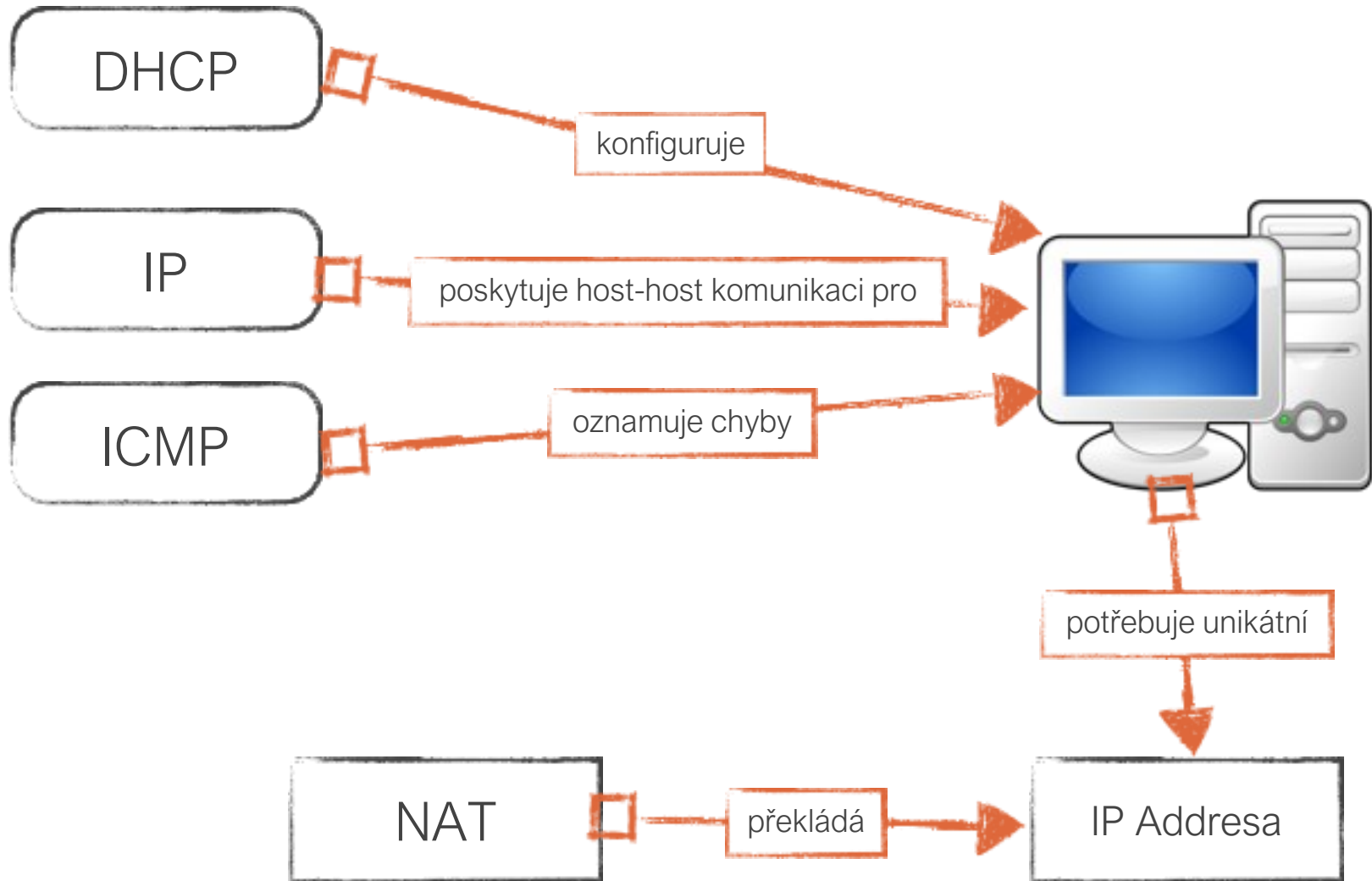
# CO vs. CL (2)



- ISO/OSI iniciativa



# Technologie

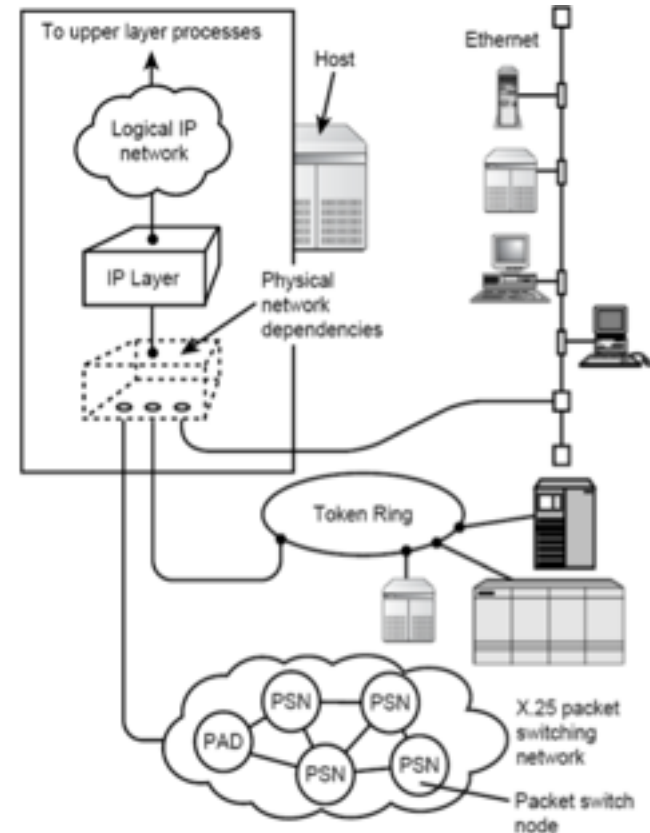


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- Protokol DHCP
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- Překlad adres, NAT

# IPv4

- Internet Protocol verze 4
- [RFC 791](#), STD [5], rok 1981
- IP pakety přenášeny rámci linkové vrstvy
  - Různé technologie linkové vrstvy (Ethernet, Token Bus, FDDI, ...)
- Quality of Service (QoS)
  - poskytuje možnost mapování požadavků na kvalitu služby pro aplikace
  - parametry pro QoS jsou předávány spolu s daty z transportní vrstvy



- IP Layer
    - Provides powerful logical abstraction
    - Hides Physical Layer dependency
  - Upper layer processes see a logical IP network
- PAD = packet assembler/disassembler

# RFC 791

[\[Docs\]](#) [\[txt|pdf\]](#) [\[Tracker\]](#) [\[Errata\]](#)

Updated by: [1349](#), [2474](#), [6864](#)

INTERNET STANDARD

Errata Exist

RFC: 791

INTERNET PROTOCOL

DARPA INTERNET PROGRAM

PROTOCOL SPECIFICATION

September 1981

prepared for

Defense Advanced Research Projects Agency  
Information Processing Techniques Office  
1400 Wilson Boulevard  
Arlington, Virginia 22209

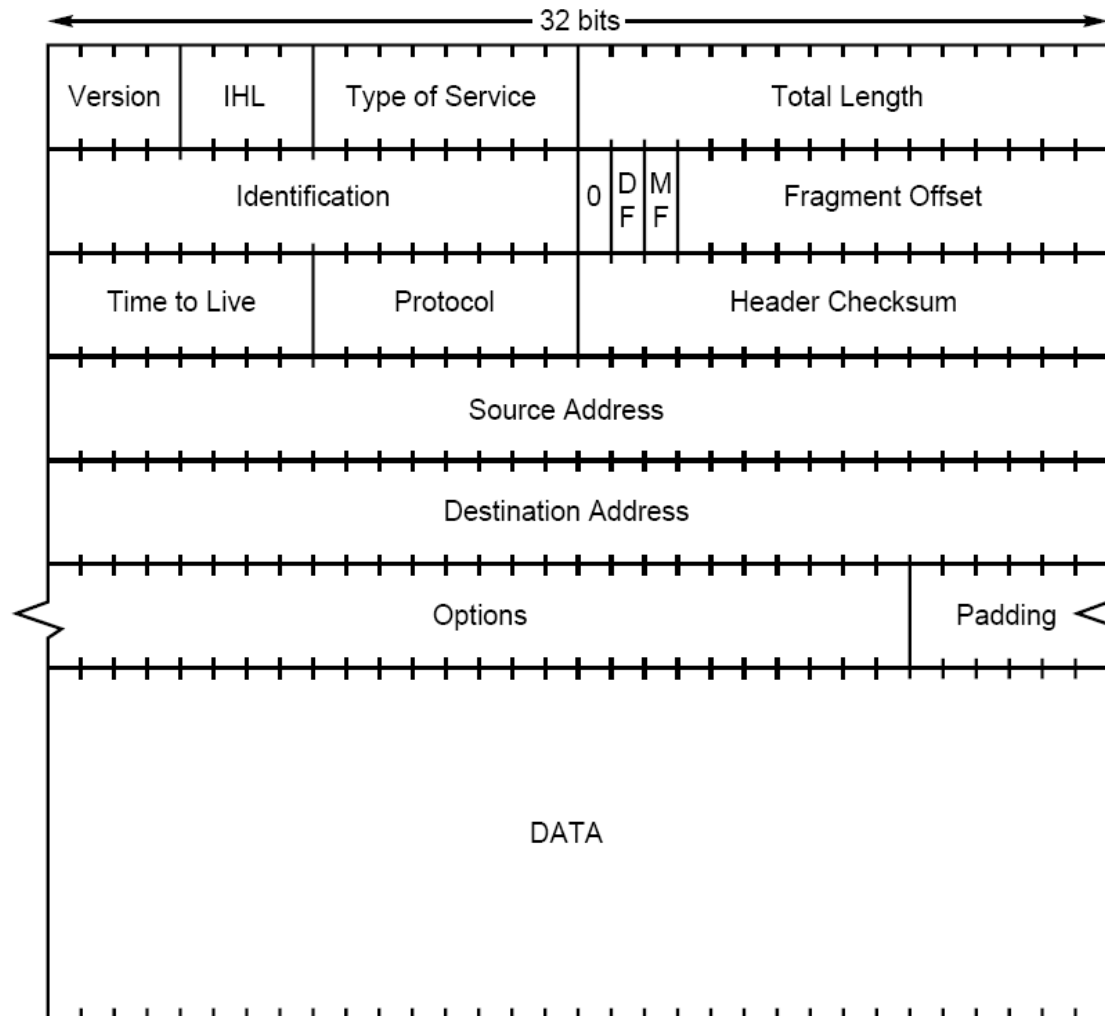
September 1981

Internet Protocol

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# IPv4 Hlavička



# IPv4 Políčka (1)

- **Verze (*Version*)**
  - verze protokolu IP, pro IPv4 hodnota 4, pro IPv6 (IPng) hodnota 6 (jiný formát rámce), hodnoty 0-3 nepoužité, 7-9 experimentální protokoly (proměnná délka adresy)
- **Délka hlavičky (*Internet Header Length, IHL*)**
  - v počtu 32 bitových slov, proměnná délka hlavičky, minimum 20 bytů, maximum 60 bytů
- **Typ služby (*Type of Service, ToS*)**
  - možnost definice požadavku kvality služby, nespecifikováno v RFC
- **Celková délka (*Total Length*)**
  - délka IP paketu (hlavička+data), maximum 65535, minimálně 576 [512 B data + 64 B hlavička]

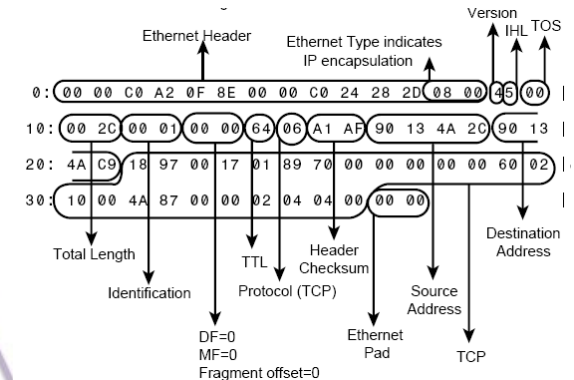
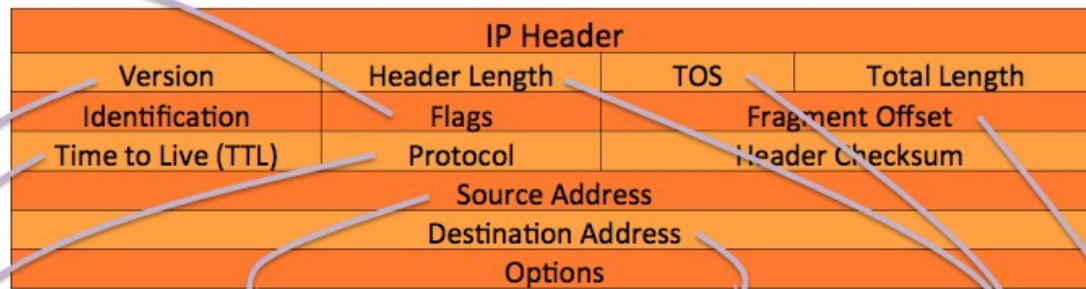
# IPv4 Políčka (2)

- **Identifikace datagramu (*Identification*)**
  - unikátní (náhodné) číslo identifikující každý datagram (většinou inkrementováno, globální čítač v OS), používá se k identifikaci fragmentace IP datagramů
- **Příznaky fragmentace a pozice fragmentu**
  - DF (Don't Fragment), MF (More Fragments)
  - Offset=počítány po 8 bytových blocích (13 bitů)
- **Životnost datagramu (*Time To Live, TTL*)**
  - původně doba v sekundách, později jako počet skoků (hop count), sníženo každým směrovačem o 1, hodnota 0 znamená vypršení doby života, datagram je zahozen, řeší zacyklení mezi směrovači
- **Protokol vyšší vrstvy (*Protocol*)**
  - identifikuje PDU protokolu, přenášeného v datech
  - 1=ICMP, 2=IGMP, 4=IP, 6=TCP, 17=UDP, 46=RSVP, 88=IGRP, 89=OSPF

# IPv4 Políčka (3)

- **Kontrolní součet hlavičky (*Header Checksum*)**
  - jedničkový doplněk součtu 16-bitových slov, bez Checksum, nutný přepočítat na každém směrovači! Proč?
- **Zdrojová adresa (*Source Address*)**
  - 32-bitová adresa hosta posílajícího datagram (host=NIC)
- **Cílová adresa (*Destination Address*)**
  - 32-bitová adresa hosta přijímacího datagram
- **Volby**
  - volitelné, bezpečnost, timestamp, record route, strict source routing

# Příklad IP Paketu



|                                                                                                          |  |
|----------------------------------------------------------------------------------------------------------|--|
| Internet Protocol Version 4, Src: 10.100.16.200 (10.100.16.200), Dst: 10.100.185.66 (10.100.185.66)      |  |
| Version: 4                                                                                               |  |
| Header length: 20 bytes                                                                                  |  |
| Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport)) |  |
| 0000 00.. = Differentiated Services Codepoint: Default (0x00)                                            |  |
| .... 00 = Explicit Congestion Notification: Not-ECT (Not ECN-Capable Transport) (0x00)                   |  |
| Total Length: 1420                                                                                       |  |
| Identification: 0x126d (4717)                                                                            |  |
| Flags: 0x02 (Don't Fragment)                                                                             |  |
| 0... .... = Reserved bit: Not set                                                                        |  |
| .1... .... = Don't fragment: Set                                                                         |  |
| ...0. .... = More fragments: Not set                                                                     |  |
| Fragment offset: 0                                                                                       |  |
| Time to live: 255                                                                                        |  |
| Protocol: TCP (6)                                                                                        |  |
| Header checksum: 0x98ad [correct]                                                                        |  |
| [Good: True]                                                                                             |  |
| [Bad: False]                                                                                             |  |
| Source: 10.100.16.200 (10.100.16.200)                                                                    |  |
| Destination: 10.100.185.66 (10.100.185.66)                                                               |  |

# Minimum Transmission Unit

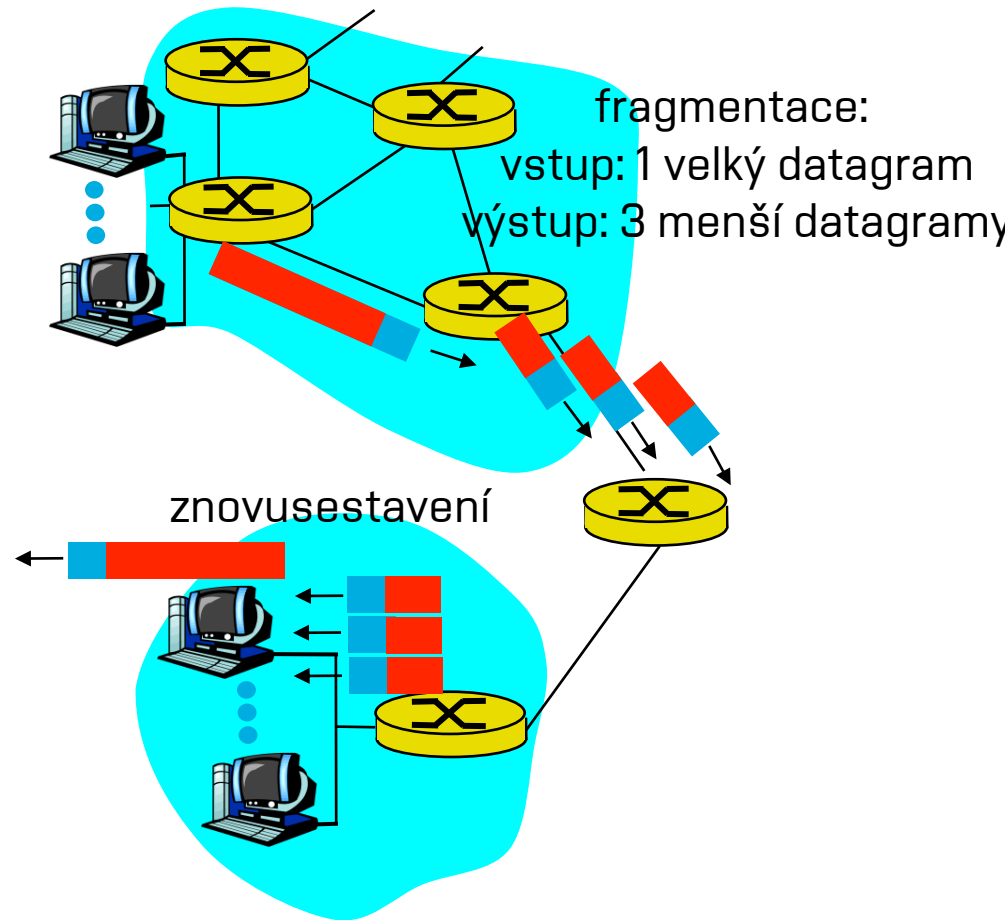
- Každá IPv4 implementace musí být schopna povinně přenést 576 bytů (celý IP datagram)
- MTU závisí na linkové vrstvě (přenosová technologie)

| Media Type    | Speed    | Frame MTU | Maximum Frame Length |
|---------------|----------|-----------|----------------------|
| Ethernet      | 10 Mbps  | 1500      | 1518                 |
| 802.3         | 10 Mbps  | 1518      | 1536                 |
| 802.3u        | 100 Mbps | 1518      | 1536                 |
| 802.3z/ab     | 1 Gbps   | 1518      | 1536                 |
| 802.3ae/an    | 10 Gbps  | 1518      | 1536                 |
| 802.3ba       | 100 Gbps | 1518?     | 1536?                |
| 802.5         | 4 Mbps   | 4528      | 4550                 |
| 802.5         | 16 Mbps  | 18173     | 18200                |
| 802.5         | 100 Mbps | 18173     | 18200                |
| 802.11g       | 54 Mbps  | 2312      | 2346                 |
| FDDI          | 100 Mbps | 4352      | 4470                 |
| Fiber Channel | 2 Gbps   | 65280     | 65280                |
| POS/OC48      | 2.5 Gbps | 9180      | 9180                 |
| ATM/AAL5      |          | 9180      | 9180                 |

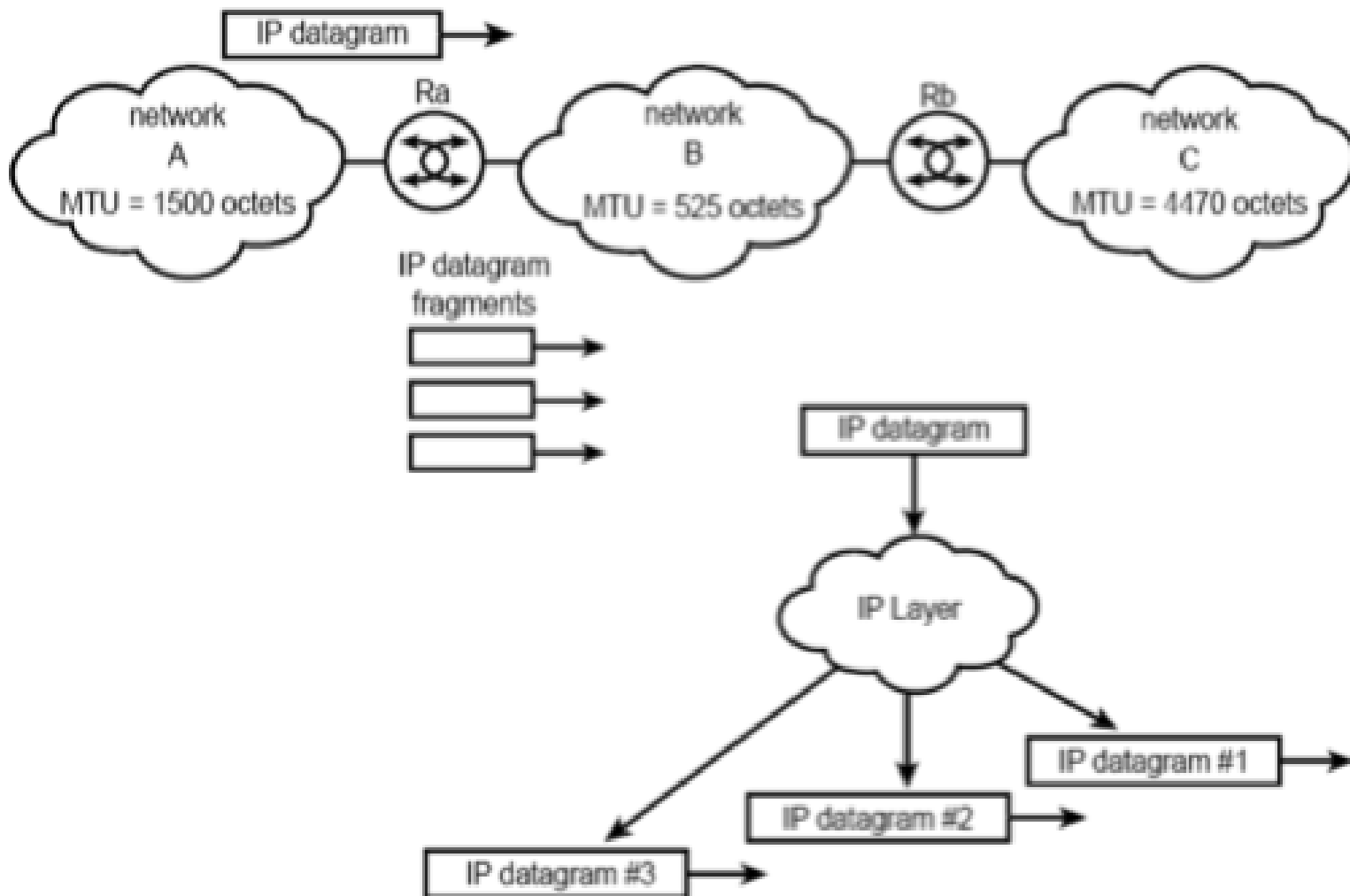
- Příklad:
  - Token Ring@4Mbps
  - $THT = 8.58 \text{ ms} * 4 \text{ Mbps} = 36000 \text{ bits} = 4500 \text{ B}$
  - $4500 - 15 \text{ (MAC header)} - 30 \text{ (Routing inf.)} - 4 \text{ (LLC header)} - 5 \text{ (SNAP header)} - 6 \text{ (MAC trailer)} = 4440 \text{ B}$

# Fragmentace

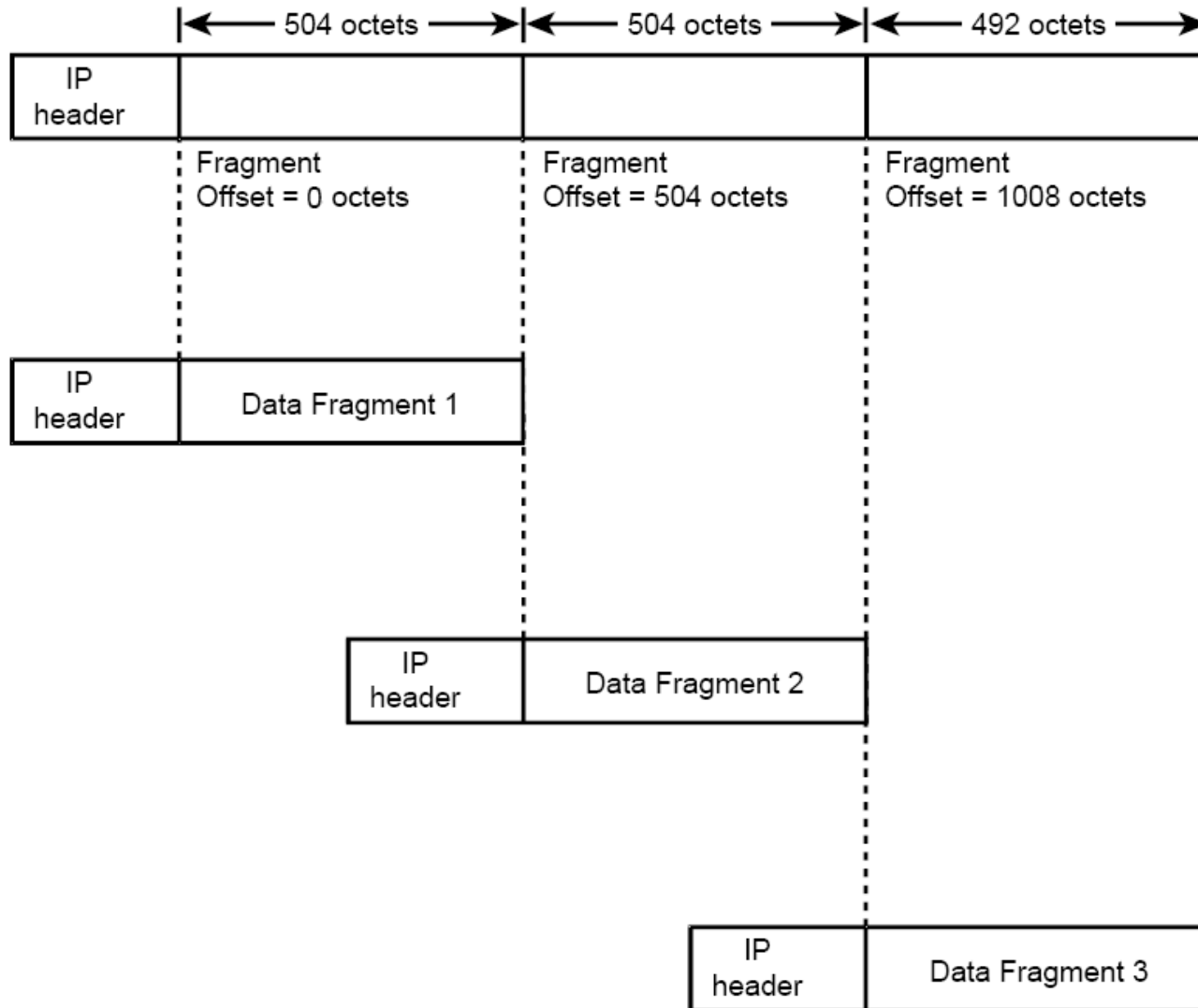
- Fyzická linka omezuje velikost přenášených dat (MTU)
  - Různé linky => různé MTU
- Pokud je v cestě linka s  $MTU < \text{velikost IP datagramu}$  a  $DF=0$ , potom směrovač fragmentuje původní IP datagram
- Datagram sestavuje až koncový host (nikoliv další směrovač v cestě)
- Při sestavení se čeká určitou dobu; pokud vyprší časovač, ICMP zpráva indikuje expiraci časovače
- Nadměrná fragmentace vede k nadměrnému přenosu a nižší efektivitě
- Path MTU Discovery RFC 1191, RFC 2923



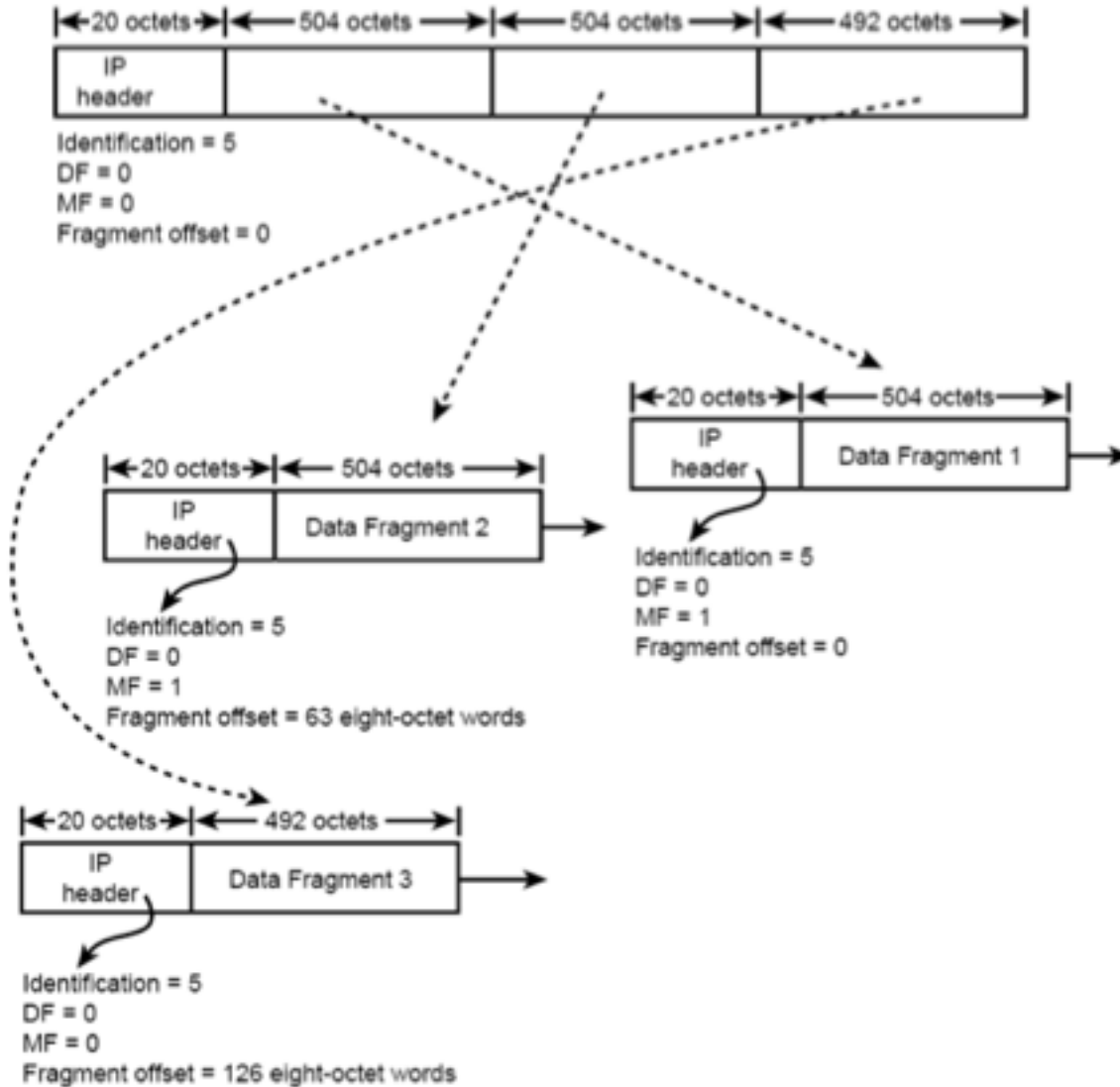
# Příklad fragmentace (1)



# Příklad fragmentace (2)

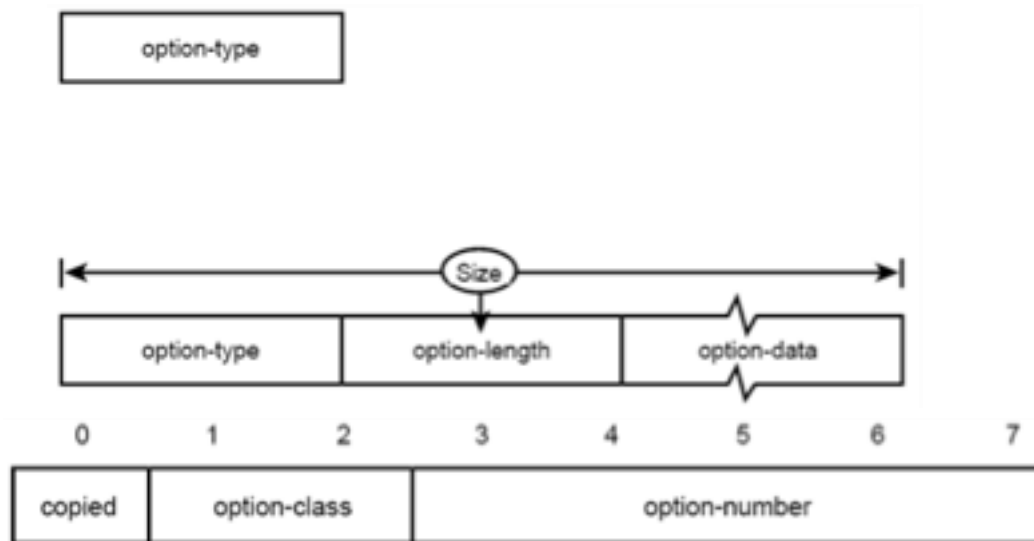


# Příklad fragmentace (3)



# Volby v IP hlavičce

- 2 formáty: single octet, variable length
- Copied: má se volba dostat do fragmetů?
- Option class: 0 Network control, 2 Debug & Measurement, zbytek reserved
- Option number: 0 End of Option List, 1 No Operation, 7 Record Route, 4 Timestamp



# Obsah

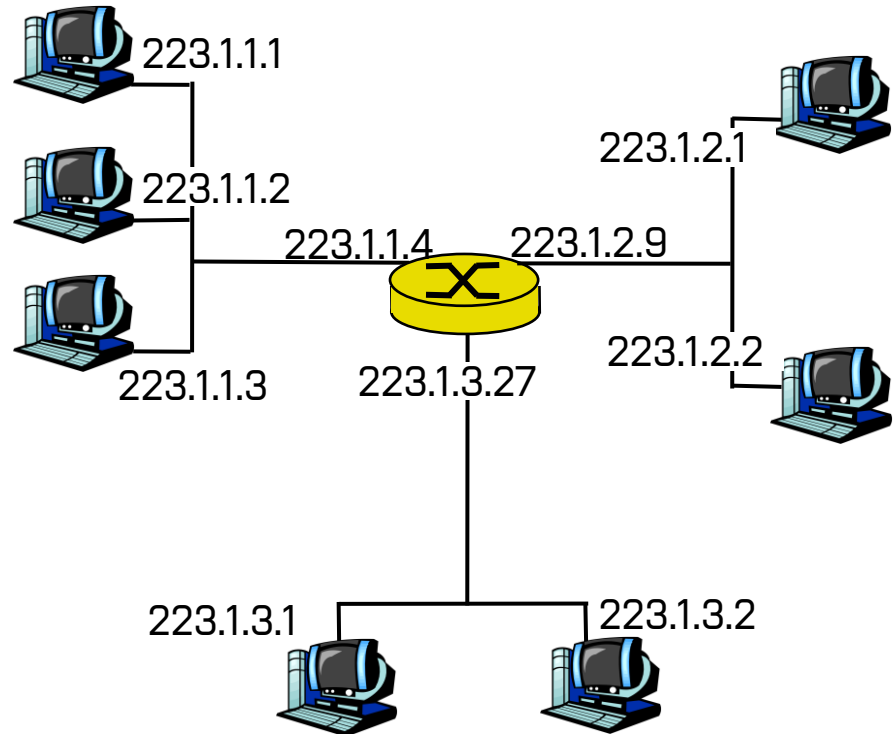
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# IPv4 adresování

- IPv4 používá adresu pevné **délky 32 bitů**
- Zápis jednotlivých oktetů 147.229.176.14
- Kolik je různých adres?
  - 0.0.0.0 a 255.255.255.255 jsou rezervované
- Adresa se skládá ze 2 částí:
  - **Identifikace sítě (NetId) + Identifikace hosta (HostId)**
  - Identifikace sítě, identifikace hosta: 147.229.0.0 adresa sítě; 0.0.176.14 adresa hosta
- Třídní adresování (**Classfull addressing**)
  - Původní koncept, máme třídy adres A-E, jasné co je adresa sítě a adresa hosta
- Beztrídní adresování (**Classless addressing**)
  - Adresa je tvořena adresou sítě, adresou podsítě a adresou hosta, hranici mezi podsítí a je určena maskou

# Proč potřebuje znát NetID?

- Síťové rozhraní (interface): spojení mezi hostem/směrovačem a fyzickým přenosovým médiem
  - Směrovač jich má obvykle více
  - Každé je identifikováno adresou
- V rámci lokální sítě/podsítě se pakety doručují bez intervence směrovače (brána, first hop router)
  - Používá se adresa linkové vrstvy



$$223.1.1.1 = \underbrace{11011111}_{223} \underbrace{00000001}_1 \underbrace{00000001}_1 \underbrace{00000001}_1$$

# Třídy adres

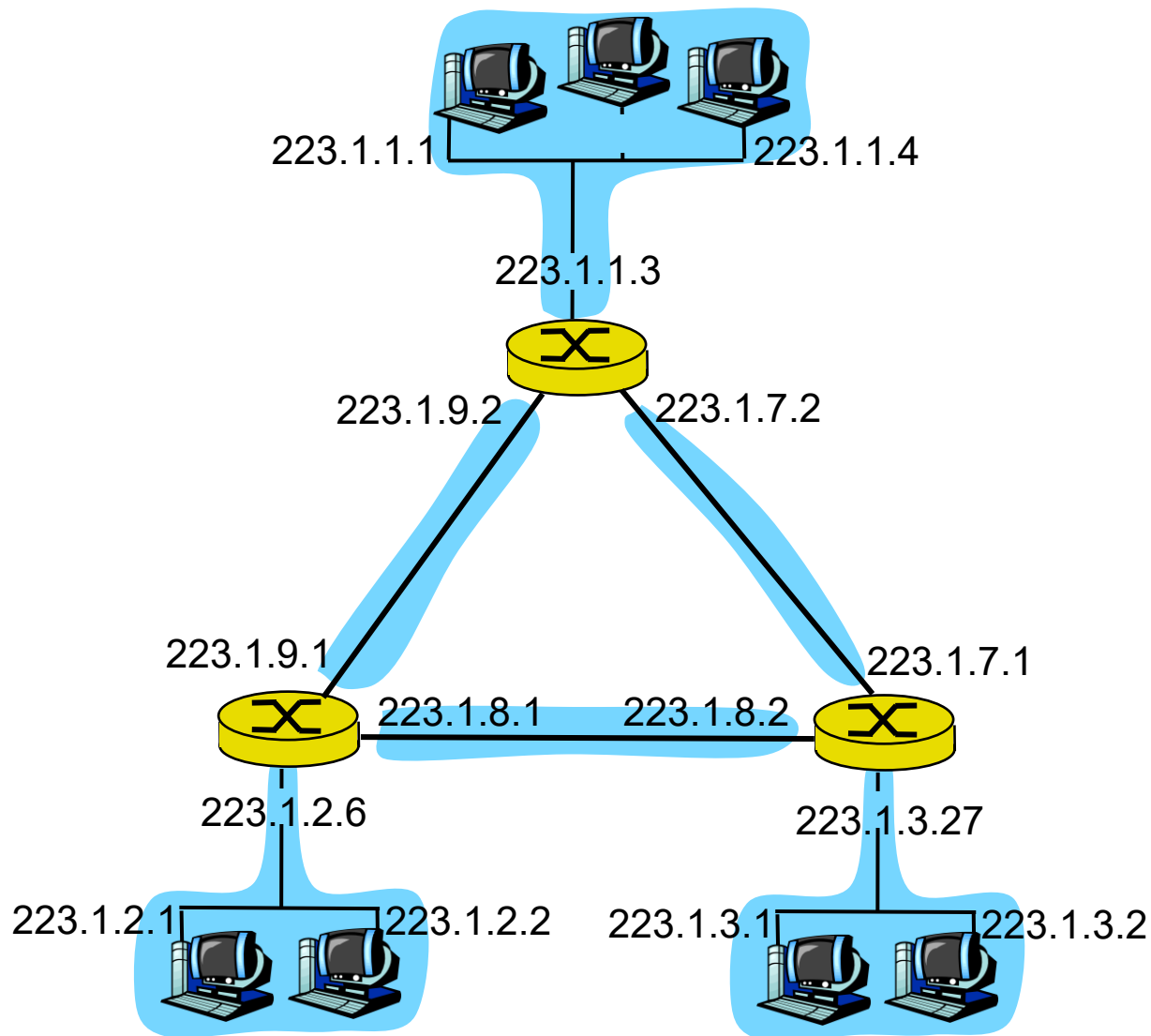
Třídy IP adres

| Třída | Začátek (bin) | 1. bajt | Standardní maska              | Bitů sítě | Bitů stanice | Síti      | Stanic v každé síti |
|-------|---------------|---------|-------------------------------|-----------|--------------|-----------|---------------------|
| A     | 0             | 0–127   | 255.0.0.0                     | 7         | 24           | 126       | 16 777 214          |
| B     | 10            | 128–191 | 255.255.0.0                   | 14        | 16           | 16384     | 65534               |
| C     | 110           | 192–223 | 255.255.255.0                 | 21        | 8            | 2 097 152 | 254                 |
| D     | 1110          | 224–239 | <i>multicast</i>              |           |              |           |                     |
| E     | 1111          | 240–255 | <i>vyhrazeno jako rezerva</i> |           |              |           |                     |

Rozsahy IP adres a masky sítě

| Třída | 1. bajt | Minimum   | Maximum         | Maska podsítě   |
|-------|---------|-----------|-----------------|-----------------|
| A     | 0–127   | 0.0.0.0   | 127.255.255.255 | 255.0.0.0       |
| B     | 128–191 | 128.0.0.0 | 191.255.255.255 | 255.255.0.0     |
| C     | 192–223 | 192.0.0.0 | 223.255.255.255 | 255.255.255.0   |
| D     | 224–239 | 224.0.0.0 | 239.255.255.255 | 255.255.255.255 |
| E     | 240–255 | 240.0.0.0 | 255.255.255.255 | -----           |

# Počet podsítí?



# Speciální adresy a typy

- Adresa obecně:
  - **Unicast** – konkrétní počítač (individuální adresa)
  - **Broadcast** - všechny počítače na síti (všesměrová adresa)
  - **Multicast** – skupina počítačů, člen skupiny může být v libovolné síti (skupinová adresa)
- **IP address ::= { network , host }**
  - { network , 0 } – “This” Network
  - { network , -1 } – Directed Broadcast
  - { -1 , -1 } – Limited Broadcast, pouze jako cílová adresa
  - { 0 , 0 } – All Zeros IP Address, pouze jako zdrojová adresa
  - { 0 , host } – IP address on This Network
  - { 127 , <any> } – Software Loopback

# Subnetting

- V průběhu 80 let RFC 917, RFC 950
- Zavedlo mechanismus podsítě
- **Maska podsítě (subnet mask)**
  - 32 bitové číslo (255.255.255.192)
  - Kde je hranice mezi adresou podsítě a adresou hosta
- Původně *all 0* nebo *all 1* v adrese podsítě nebyly povolené, později ano viz. RFC 1812

# Subnetting síť B třídy

| <i>Number of Bits<br/>in Network<br/>Addresses<br/>Prefix</i> | <i>Subnet Mask</i> | <i>Number of<br/>Usable Subnet</i> | <i>Number of Usable Host<br/>Addresses, Per Subnet</i> |
|---------------------------------------------------------------|--------------------|------------------------------------|--------------------------------------------------------|
| 2                                                             | 255.255.192.0      | 4                                  | 16,382                                                 |
| 3                                                             | 255.255.224.0      | 8                                  | 8,190                                                  |
| 4                                                             | 255.255.240.0      | 16                                 | 4,094                                                  |
| 5                                                             | 255.255.248.0      | 32                                 | 2,046                                                  |
| 6                                                             | 255.255.252.0      | 64                                 | 1,022                                                  |
| 7                                                             | 255.255.254.0      | 128                                | 510                                                    |
| 8                                                             | 255.255.255.0      | 256                                | 254                                                    |
| 9                                                             | 255.255.255.128    | 512                                | 126                                                    |
| 10                                                            | 255.255.255.192    | 1,024                              | 62                                                     |
| 11                                                            | 255.255.255.224    | 2,048                              | 30                                                     |
| 12                                                            | 255.255.255.240    | 4,096                              | 14                                                     |
| 13                                                            | 255.255.255.248    | 8,192                              | 6                                                      |
| 14                                                            | 255.255.255.252    | 16,384                             | 2                                                      |

# Příklad subnetting

- Síť 193.168.125.0/24 rozdělte na 8 podsítí /27!*

| <i>Network #</i> | <i>Binary Address</i>                | <i>Decimal Address</i> |
|------------------|--------------------------------------|------------------------|
| Base             | 11000001.10101000.01111101.00000000  | 193.168.125.0          |
| Subnet 0         | 11000001.10101000.01111101.000-00000 | 193.168.125.0          |
| Subnet 1         | 11000001.10101000.01111101.001-00000 | 193.168.125.32         |
| Subnet 2         | 11000001.10101000.01111101.010-00000 | 193.168.125.64         |
| Subnet 3         | 11000001.10101000.01111101.011-00000 | 193.168.125.96         |
| Subnet 4         | 11000001.10101000.01111101.100-00000 | 193.168.125.128        |
| Subnet 5         | 11000001.10101000.01111101.101-00000 | 193.168.125.160        |
| Subnet 6         | 11000001.10101000.01111101.110-00000 | 193.168.125.192        |
| Subnet 7         | 11000001.10101000.01111101.111-00000 | 193.168.125.224        |

# Příklady ①

- Jaká je adresa sítě, do které patří host s adresou 172.16.2.160/26?*

|                 | Network  |          | Subnet   | Host     |
|-----------------|----------|----------|----------|----------|
| 172.16.2.160    | 10101100 | 00010000 | 00000010 | 10100000 |
| 255.255.255.192 | 11111111 | 11111111 | 11111111 | 11000000 |

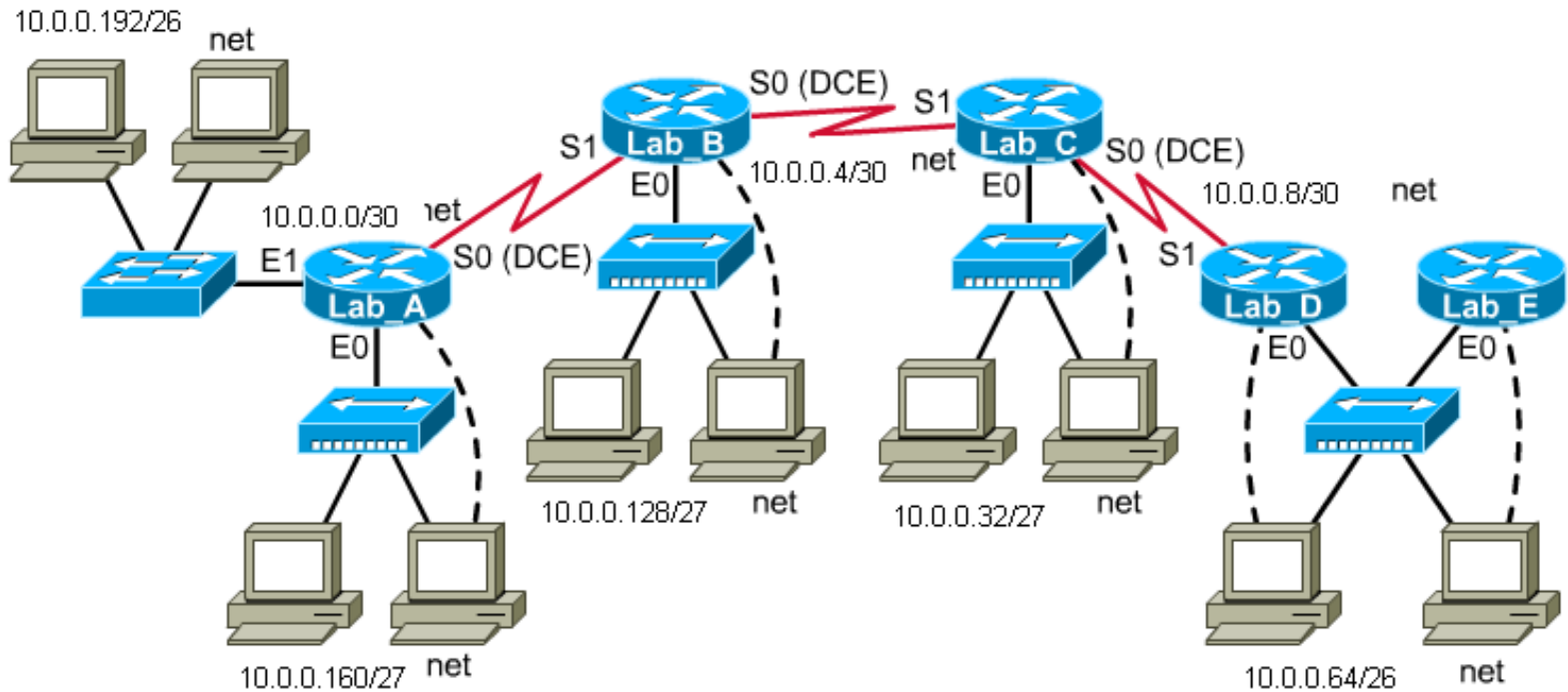
# Příklady ②

- Spočítejte adresu sítě, broadcastovou adresu a první + poslední použitelnou hostovskou adresu, pokud vám je zadána IP 174.24.4.176/26!*

|                 | 174      | 24       | 4        | 176      |           |
|-----------------|----------|----------|----------|----------|-----------|
| 174.24.4.176    | 10101110 | 00011000 | 00000100 | 10110000 | Host      |
| 255.255.255.192 | 11111111 | 11111111 | 11111111 | 11000000 | Mask      |
|                 |          |          |          |          | Subnet    |
|                 |          |          |          |          | Broadcast |
|                 |          |          |          |          | First     |
|                 |          |          |          |          | Last      |

# Příklady ③

- Rozdělte síť 10.0.0.0/24 na podsítě tak, aby platilo:
  - A, B, C spojnice pro 2 hosty / D, E, F pro 30 hostů / G, H pro 62 hostů



```
Router Name - Lab_A
Router Type - 2514
E0 =10.0.0.161
E1 =10.0.0.193
S0 =10.0.0.1
```

```
Router Name - Lab_B
Router Type - 2503
E0 =10.0.0.129
S0 =10.0.0.5
S1 =10.0.0.2
```

```
Router Name - Lab_C
Router Type - 2503
E0 = 10.0.0.33
S0 = 10.0.0.9
S1 = 10.0.0.6
```

```
Router Name - Lab_D
Router Type - 2501
E0 = 10.0.0.65
S1 = 10.0.0.10
```

```
Router Name - Lab_E
Router Type - 2501
E0 = 10.0.0.66
```

# Pomůcky (1)

| /slash | # Hosts    | Netmask         | Wildcard      |
|--------|------------|-----------------|---------------|
| /30    | 4          | 255.255.255.252 | 0.0.0.3       |
| /29    | 8          | 255.255.255.248 | 0.0.0.7       |
| /28    | 16         | 255.255.255.240 | 0.0.0.15      |
| /27    | 32         | 255.255.255.224 | 0.0.0.31      |
| /26    | 64         | 255.255.255.192 | 0.0.0.63      |
| /25    | 128        | 255.255.255.128 | 0.0.0.127     |
| /24    | 256        | 255.255.255.0   | 0.0.0.255     |
| /23    | 512        | 255.255.254.0   | 0.0.1.255     |
| /22    | 1,024      | 255.255.252.0   | 0.0.3.255     |
| /21    | 2,048      | 255.255.248.0   | 0.0.7.255     |
| /20    | 4,096      | 255.255.240.0   | 0.0.15.255    |
| /19    | 8,192      | 255.255.224.0   | 0.0.32.255    |
| /18    | 16,384     | 255.255.192.0   | 0.0.63.255    |
| /17    | 32,768     | 255.255.128.0   | 0.0.127.255   |
| /16    | 65,536     | 255.255.0.0     | 0.0.255.255   |
| /15    | 131,072    | 255.254.0.0     | 0.1.255.255   |
| /14    | 262,144    | 255.252.0.0     | 0.3.255.255   |
| /13    | 524,288    | 255.248.0.0     | 0.7.255.255   |
| /12    | 1,048,576  | 255.240.0.0     | 0.15.255.255  |
| /11    | 2,097,152  | 255.224.0.0     | 0.31.255.255  |
| /10    | 4,194,304  | 255.192.0.0     | 0.63.255.255  |
| /9     | 8,388,608  | 255.128.0.0     | 0.127.255.255 |
| /8     | 16,777,216 | 255.0.0.0       | 0.255.255.255 |

| Subnet mask quick reference |           |           |                 |            |             |             |               |
|-----------------------------|-----------|-----------|-----------------|------------|-------------|-------------|---------------|
| Host Bit length             | math      | Max hosts | Subnet mask     | Mask octet | Binary mask | Mask length | Subnet length |
| 0                           | $2^0=$    | 1         | 255.255.255.255 | 4          | 11111111    | 32          | 0             |
| 1                           | $2^1=$    | 2         | 255.255.255.254 | 4          | 11111110    | 31          | 1             |
| 2                           | $2^2=$    | 4         | 255.255.255.252 | 4          | 11111100    | 30          | 2             |
| 3                           | $2^3=$    | 8         | 255.255.255.248 | 4          | 11111000    | 29          | 3             |
| 4                           | $2^4=$    | 16        | 255.255.255.240 | 4          | 11110000    | 28          | 4             |
| 5                           | $2^5=$    | 32        | 255.255.255.224 | 4          | 11100000    | 27          | 5             |
| 6                           | $2^6=$    | 64        | 255.255.255.192 | 4          | 11000000    | 26          | 6             |
| 7                           | $2^7=$    | 128       | 255.255.255.128 | 4          | 10000000    | 25          | 7             |
| 8                           | $2^8=$    | 256       | 255.255.255.0   | 3          | 11111111    | 24          | 8             |
| 9                           | $2^9=$    | 512       | 255.255.254.0   | 3          | 11111110    | 23          | 9             |
| 10                          | $2^{10}=$ | 1024      | 255.255.252.0   | 3          | 11111100    | 22          | 10            |
| 11                          | $2^{11}=$ | 2048      | 255.255.248.0   | 3          | 11111000    | 21          | 11            |
| 12                          | $2^{12}=$ | 4096      | 255.255.240.0   | 3          | 11110000    | 20          | 12            |
| 13                          | $2^{13}=$ | 8192      | 255.255.224.0   | 3          | 11100000    | 19          | 13            |
| 14                          | $2^{14}=$ | 16384     | 255.255.192.0   | 3          | 11000000    | 18          | 14            |
| 15                          | $2^{15}=$ | 32768     | 255.255.128.0   | 3          | 10000000    | 17          | 15            |
| 16                          | $2^{16}=$ | 65536     | 255.255.0.0     | 2          | 11111111    | 16          | 16            |
| 17                          | $2^{17}=$ | 131072    | 255.254.0.0     | 2          | 11111110    | 15          | 17            |
| 18                          | $2^{18}=$ | 262144    | 255.252.0.0     | 2          | 11111100    | 14          | 18            |
| 19                          | $2^{19}=$ | 524288    | 255.248.0.0     | 2          | 11111000    | 13          | 19            |
| 20                          | $2^{20}=$ | 1048576   | 255.240.0.0     | 2          | 11110000    | 12          | 20            |
| 21                          | $2^{21}=$ | 2097152   | 255.224.0.0     | 2          | 11100000    | 11          | 21            |
| 22                          | $2^{22}=$ | 4194304   | 255.192.0.0     | 2          | 11000000    | 10          | 22            |
| 23                          | $2^{23}=$ | 8388608   | 255.128.0.0     | 2          | 10000000    | 9           | 23            |
| 24                          | $2^{24}=$ | 16777216  | 255.0.0.0       | 1          | 11111111    | 8           | 24            |

<http://c128.com/sites/default/files/field/image/netmask.JPG>

<https://s-media-cache-ak0.pinimg.com/564x/9b/df/d9/9bdf9263fbede0be625266acf9be292.jpg>

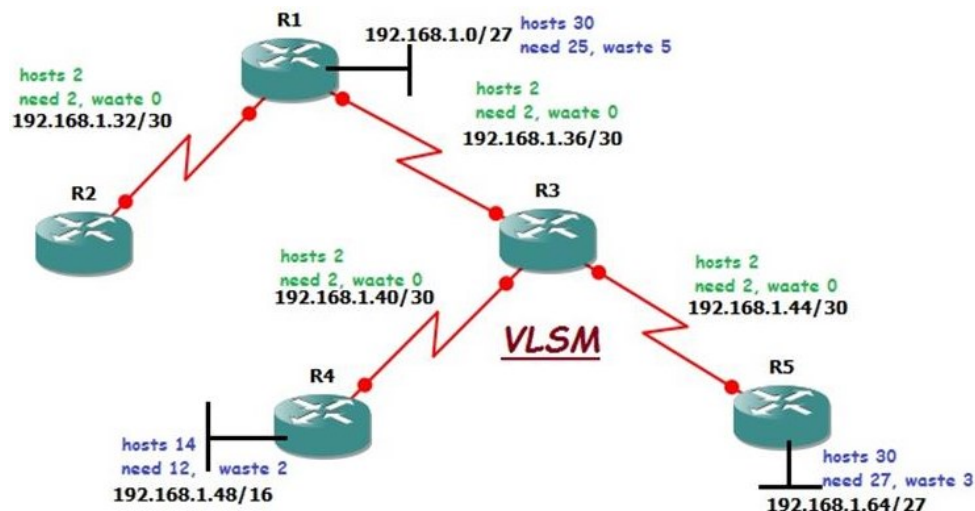
# Pomůcky (2)

[http://www.ic-ims.com/informaticke\\_mreze/vism\\_eng/vism-and-subnetting-vism-tablica.gif](http://www.ic-ims.com/informaticke_mreze/vism_eng/vism-and-subnetting-vism-tablica.gif)

|      | / 25<br>255.255.255.128 | / 26<br>255.255.255.192 | / 27<br>255.255.255.224 | / 28<br>255.255.255.240 | / 29<br>255.255.255.248 | / 30<br>255.255.255.252 |
|------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| .0   |                         |                         |                         | .0 (.1-.14)             | .0 (.1-.6)              | .0 (.1-.2)              |
| .4   |                         |                         |                         | .8 (.9-.14)             | .8 (.9-.14)             | .8 (.9-.10)             |
| .8   |                         |                         |                         | .16 (.13-.14)           | .16 (.13-.14)           | .16 (.13-.14)           |
| .12  |                         |                         | .0 (.1-.30)             | .16 (.17-.22)           | .16 (.17-.22)           | .16 (.17-.18)           |
| .16  |                         |                         |                         | .20 (.21-.22)           | .24 (.25-.30)           | .24 (.25-.26)           |
| .20  |                         |                         |                         | .24 (.25-.30)           | .28 (.29-.30)           | .28 (.29-.30)           |
| .24  |                         |                         |                         | .32 (.33-.34)           | .36 (.37-.38)           | .36 (.37-.38)           |
| .28  |                         | .0 (.1-.62)             |                         | .40 (.41-.42)           | .44 (.45-.46)           | .44 (.45-.46)           |
| .32  |                         |                         | .32 (.33-.62)           | .48 (.49-.54)           | .52 (.53-.54)           | .52 (.53-.54)           |
| .36  |                         |                         |                         | .56 (.57-.62)           | .60 (.61-.62)           | .60 (.61-.62)           |
| .40  |                         |                         |                         | .64 (.65-.70)           | .68 (.69-.70)           | .64 (.65-.66)           |
| .44  |                         |                         |                         | .72 (.73-.78)           | .76 (.77-.78)           | .72 (.73-.74)           |
| .48  |                         |                         |                         | .80 (.81-.86)           | .84 (.85-.86)           | .80 (.81-.82)           |
| .52  |                         |                         |                         | .88 (.89-.94)           | .92 (.93-.94)           | .88 (.89-.90)           |
| .56  |                         |                         |                         | .96 (.97-.102)          | .100 (.101-.102)        | .96 (.97-.98)           |
| .60  |                         |                         |                         | .104 (.105-.106)        | .108 (.109-.110)        | .104 (.105-.106)        |
| .64  |                         |                         |                         | .112 (.113-.114)        | .116 (.117-.118)        | .112 (.113-.114)        |
| .68  |                         |                         |                         | .120 (.121-.122)        | .124 (.125-.126)        | .120 (.121-.122)        |
| .72  |                         |                         |                         | .128 (.129-.134)        | .132 (.133-.134)        | .128 (.129-.130)        |
| .76  |                         |                         |                         | .136 (.137-.142)        | .140 (.141-.142)        | .136 (.137-.138)        |
| .80  |                         |                         |                         | .144 (.145-.150)        | .148 (.149-.150)        | .144 (.145-.146)        |
| .84  |                         |                         |                         | .152 (.153-.154)        | .156 (.155-.156)        | .148 (.149-.152)        |
| .88  |                         |                         |                         | .160 (.161-.166)        | .164 (.165-.166)        | .156 (.155-.156)        |
| .92  |                         |                         |                         | .168 (.169-.174)        | .172 (.173-.174)        | .164 (.165-.166)        |
| .96  |                         |                         |                         | .176 (.177-.182)        | .180 (.181-.182)        | .172 (.173-.174)        |
| .100 |                         |                         |                         | .184 (.185-.190)        | .188 (.189-.190)        | .180 (.181-.182)        |
| .104 |                         |                         |                         | .192 (.193-.198)        | .196 (.197-.198)        | .184 (.185-.186)        |
| .108 |                         |                         |                         | .200 (.201-.202)        | .204 (.205-.206)        | .192 (.193-.194)        |
| .112 |                         |                         |                         | .208 (.209-.214)        | .212 (.213-.214)        | .200 (.201-.202)        |
| .116 |                         |                         |                         | .216 (.217-.222)        | .220 (.221-.222)        | .208 (.209-.210)        |
| .120 |                         |                         |                         | .224 (.225-.230)        | .228 (.229-.230)        | .216 (.217-.218)        |
| .124 |                         |                         |                         | .232 (.233-.238)        | .236 (.237-.238)        | .224 (.225-.226)        |
| .128 |                         |                         |                         | .240 (.241-.246)        | .244 (.245-.246)        | .232 (.233-.234)        |
| .132 |                         |                         |                         | .248 (.249-.250)        | .252 (.253-.254)        | .240 (.241-.242)        |
| .136 |                         |                         |                         |                         |                         | .248 (.249-.250)        |
| .140 |                         |                         |                         |                         |                         | .252 (.253-.254)        |
| .144 |                         |                         |                         |                         |                         |                         |
| .148 |                         |                         |                         |                         |                         |                         |
| .152 |                         |                         |                         |                         |                         |                         |
| .156 |                         |                         |                         |                         |                         |                         |
| .160 |                         |                         |                         |                         |                         |                         |
| .164 |                         |                         |                         |                         |                         |                         |
| .168 |                         |                         |                         |                         |                         |                         |
| .172 |                         |                         |                         |                         |                         |                         |
| .176 |                         |                         |                         |                         |                         |                         |
| .180 |                         |                         |                         |                         |                         |                         |
| .184 |                         |                         |                         |                         |                         |                         |
| .188 |                         |                         |                         |                         |                         |                         |
| .192 |                         |                         |                         |                         |                         |                         |
| .196 |                         |                         |                         |                         |                         |                         |
| .200 |                         |                         |                         |                         |                         |                         |
| .204 |                         |                         |                         |                         |                         |                         |
| .208 |                         |                         |                         |                         |                         |                         |
| .212 |                         |                         |                         |                         |                         |                         |
| .216 |                         |                         |                         |                         |                         |                         |
| .220 |                         |                         |                         |                         |                         |                         |
| .224 |                         |                         |                         |                         |                         |                         |
| .228 |                         |                         |                         |                         |                         |                         |
| .232 |                         |                         |                         |                         |                         |                         |
| .236 |                         |                         |                         |                         |                         |                         |
| .240 |                         |                         |                         |                         |                         |                         |
| .244 |                         |                         |                         |                         |                         |                         |
| .248 |                         |                         |                         |                         |                         |                         |
| .252 |                         |                         |                         |                         |                         |                         |

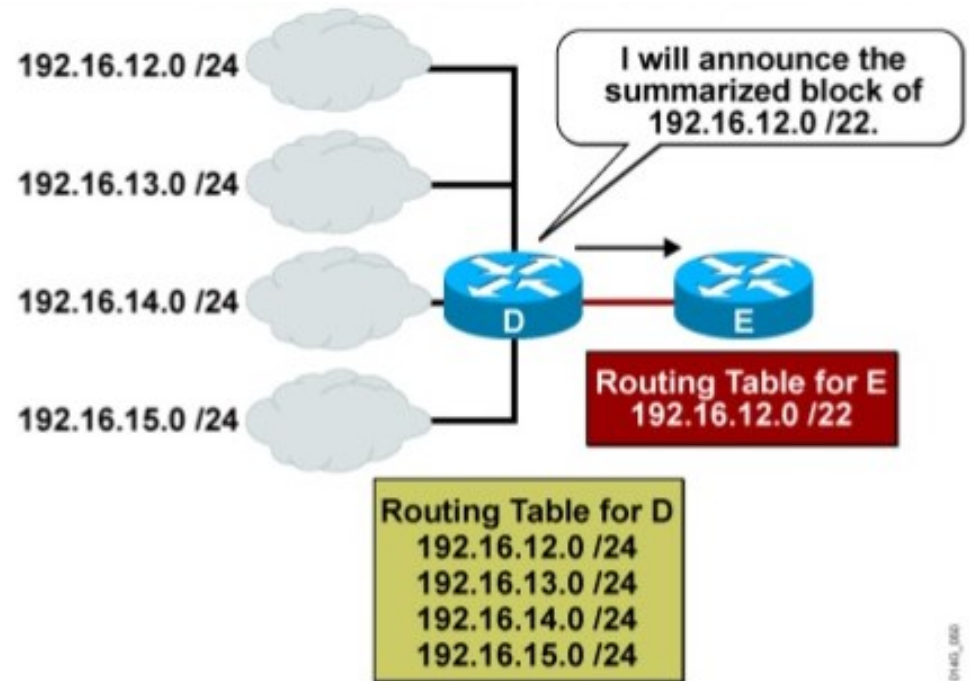
# VLSM

- Variable Length Subnet Mask (v roce 1987, RFC 1009)
- Efektivnější využití adresového prostoru
- Extended network prefix
  - 147.229.176.14/23
  - 23 počet bitů masky 255.255.254.0
- Adresa bez masky nedává smysl
- „Přidávání bitů zprava“



# CIDR

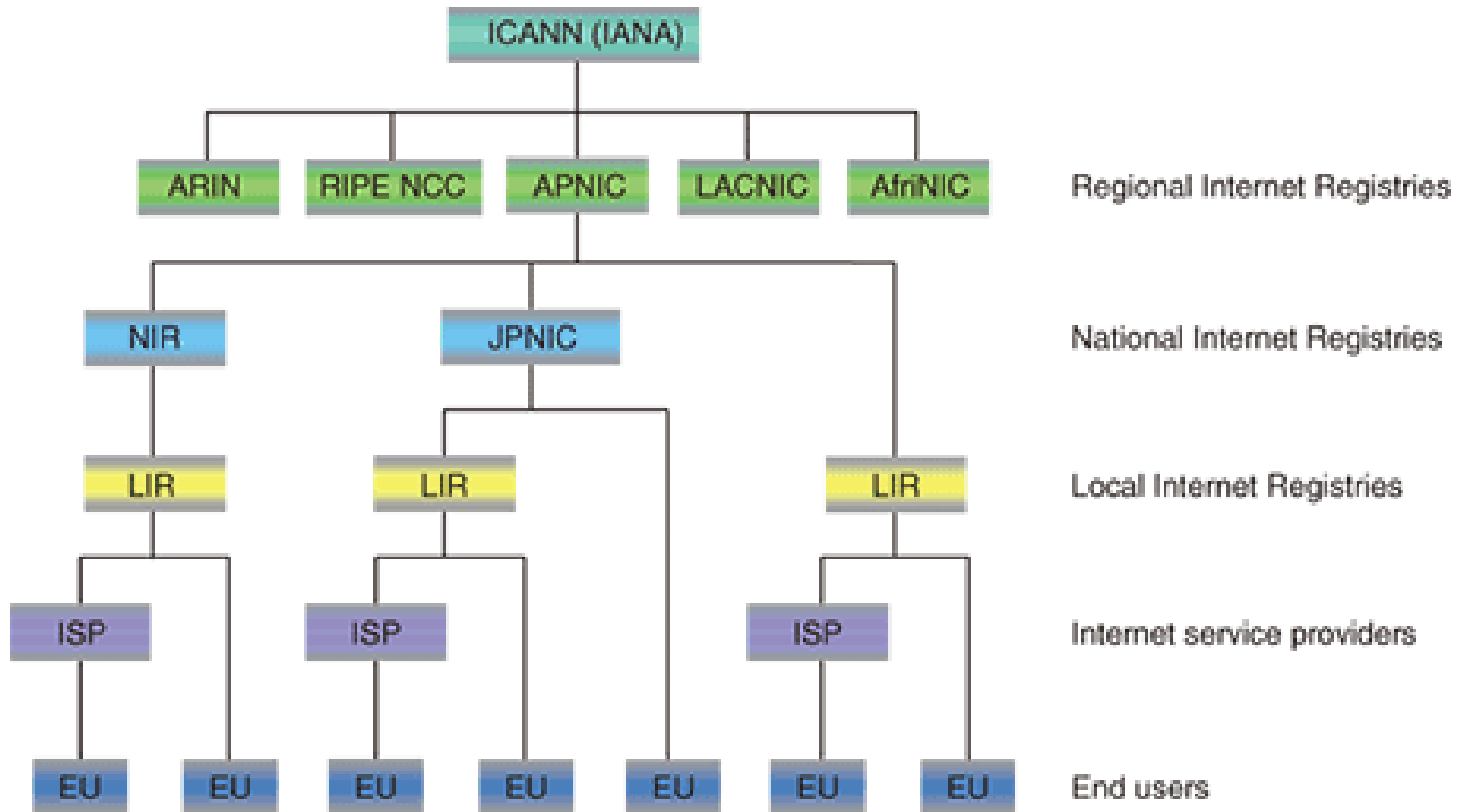
- Classless Interdomain Routing
- V roce 1992
  - Téměř vyčerpání B třídy adres
  - Směrovací tabulky příliš velké
- V roce 1993
  - Agregace adres pro směrování
  - Supernetting
  - Eliminace classfull adresování
  - RFC 1517 až 1520
  - „Přidávání bitů zleva“
- RFC 1918
  - Malá úprava: třída A je teď identifikována jako x/8, B jako x.y/16 ...



# Obsah

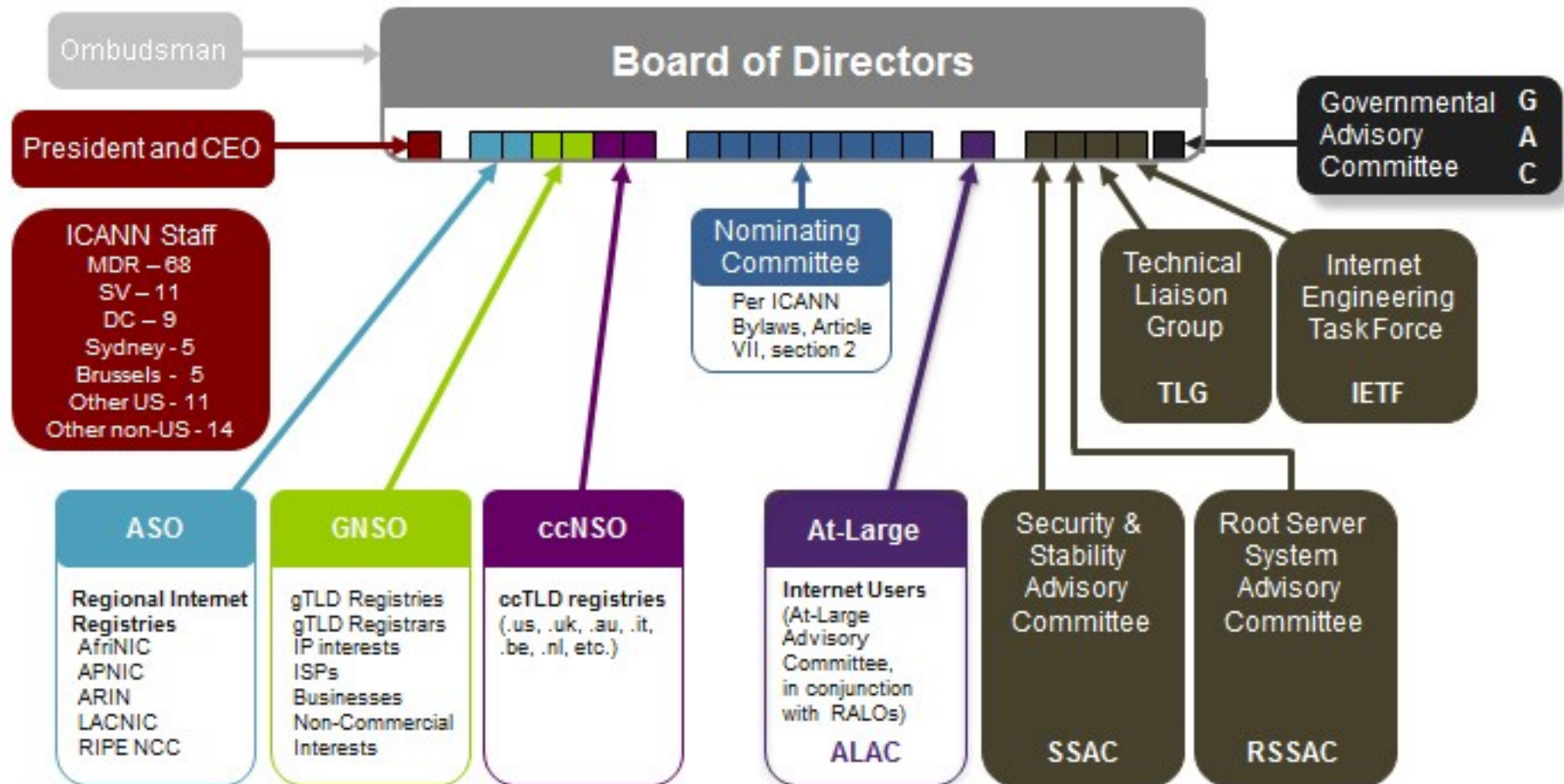
- Služby a funkce síťové vrstvy
- Protokol IPv4
  - Funkce, formát
  - Fragmentace a znovusestavení
  - Adresování v IPv4
- Správa adresního prostoru
- Protokol ICMP
  - Funkce, formát
  - Příklady služeb využívající ICMP
- Protokol DHCP
  - Funkce, formát
- Překlad adres, NAT

# Hráči



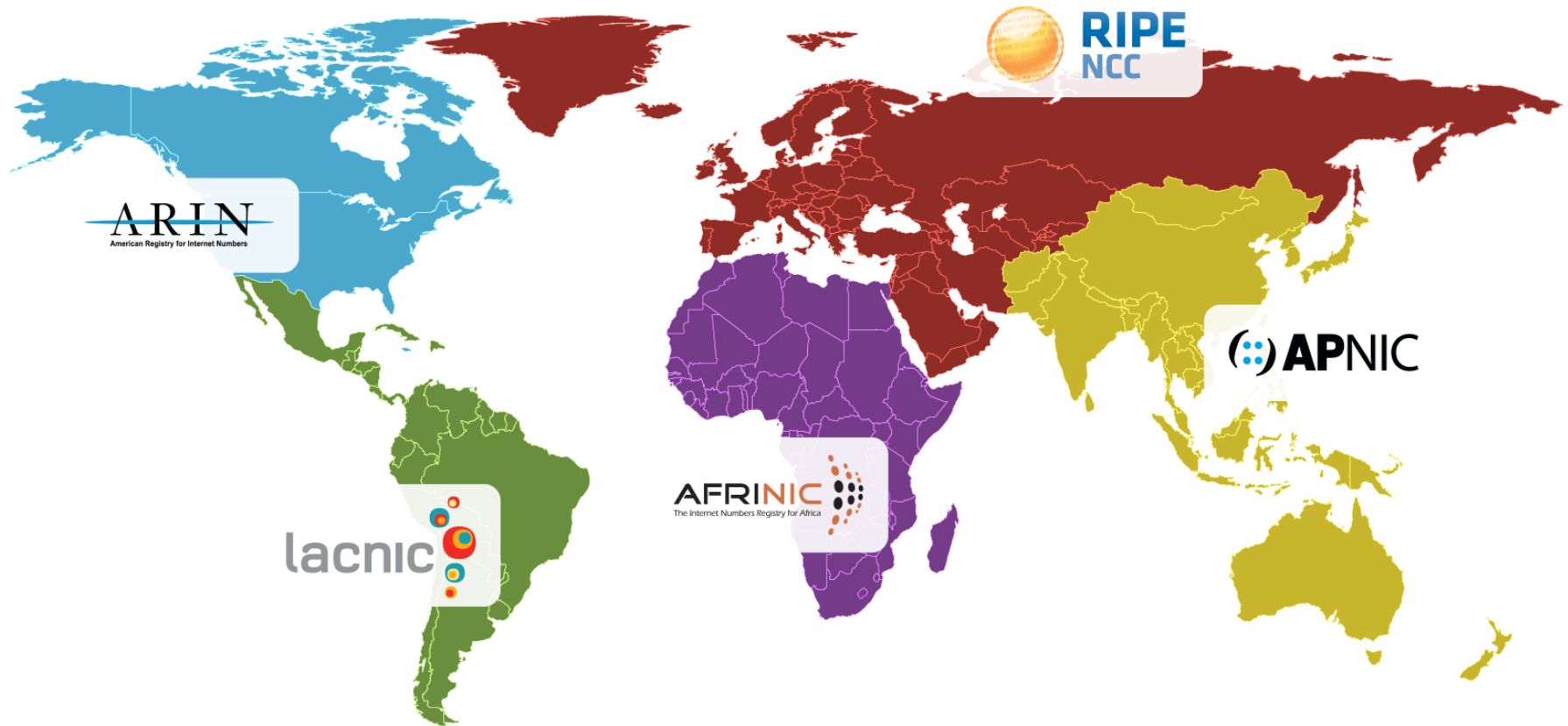
- [https://www.internetsociety.org/wp-content/uploads/2016/05/IANA\\_Timeline\\_20170117.pdf](https://www.internetsociety.org/wp-content/uploads/2016/05/IANA_Timeline_20170117.pdf)

# ICANN

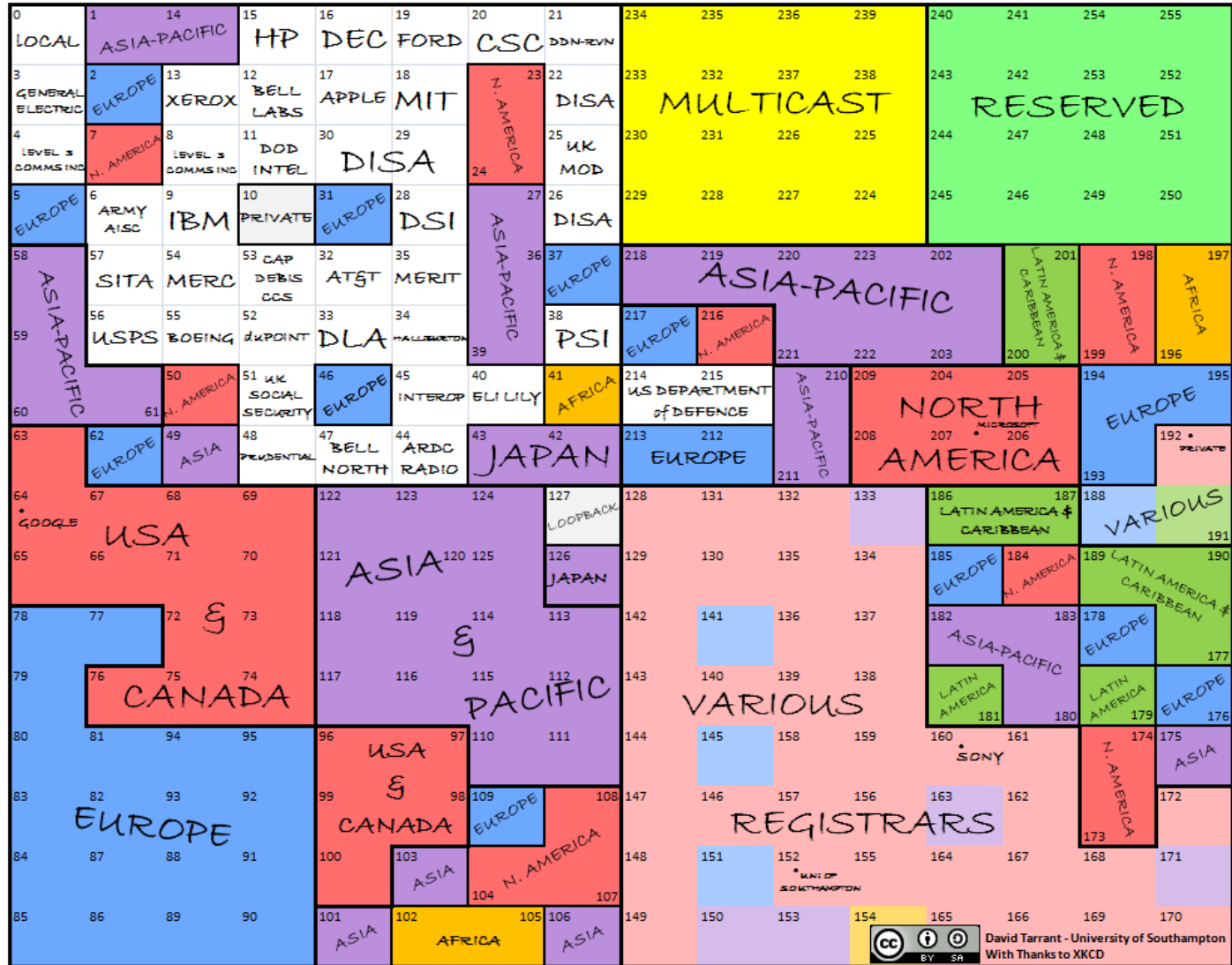


# IPv4 adresní prostor

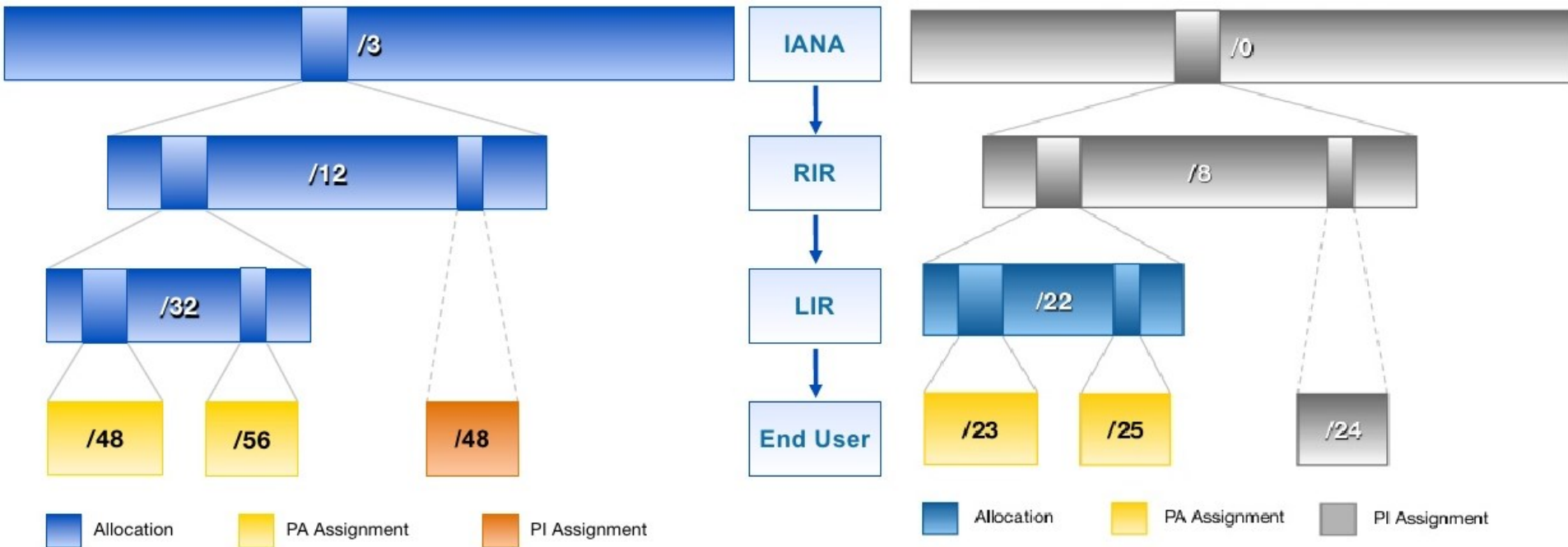
- O rozdělování adres se starají nadnárodní registrátoři



# IPv4 adresní prostor



# Delegace IP prostoru



# IPv4 sítě na páteři

- <https://ipv4.potaroo.net/> a <http://bgp.potaroo.net/as6447/>
- <http://routeserver.org/>

```
route-views.routeviews.org - PuTTY
BGP router identifier 128.223.51.103, local AS number 6447
BGP table version is 68135525, main routing table version 68135525
787893 network entries using 195397464 bytes of memory
23934128 path entries using 2872095360 bytes of memory
3616548/130500 BGP path/bestpath attribute entries using 896903904 bytes of memory
3346798 BGP AS-PATH entries using 169530898 bytes of memory
3 BGP ATTR_SET entries using 120 bytes of memory
101436 BGP community entries using 12157590 bytes of memory
859 BGP extended community entries using 55066 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 4146140282 total bytes of memory
BGP activity 3073719/2217489 prefixes, 218184206/192401772 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ  OutQ Up/Down  State/PfxRcd
4.69.184.193  4        3356      0       0        1    0    0 6w4d    Active
5.101.110.2   4        14061     0       0        1    0    0 never   Active
12.0.1.63     4        7018 33391166 129913 68135543 0    0 11w5d   724660
37.139.139.0  4        57866 1806957  4506 68135543 0    0 1d10h   729532
64.71.137.241 4        6939 18016190 129896 68135543 0    0 11w5d   754266
66.59.190.221 4        6539      0       0        1    0    0 never   Active
66.110.0.86   4        6453      0       0        1    0    0 never   Active
66.185.128.48 4        1668      0       0        1    0    0 never   Active
69.31.111.244 4        4436      0       0        1    0    0 never   Active
80.241.176.31 4        20771     0       0        1    0    0 never   Idle
89.149.178.10 4        3257      0       0        1    0    0 4d13h   Active
91.218.184.60 4        49788 27264246 97713 68135543 0    0 11w5d   731188
94.142.247.3  4        8283 172221593 10241 68135543 0    0 1w1d    732708
95.85.0.2     4        14061     0       0        1    0    0 never   Active
--More--
```

Date

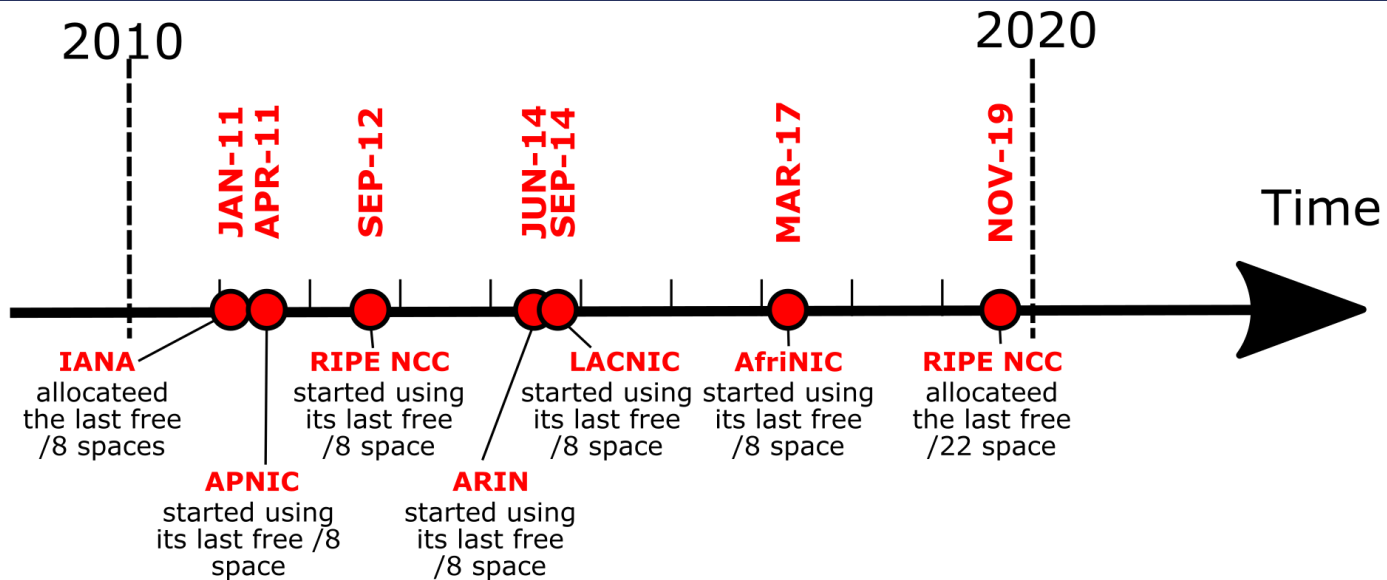
# Veřejné a privátní adresy

- Pokud síť není připojená k Internetu, můžeme použít libovolné adresy
- RFC 1597
  - definuje privátní adresy (neroutovatelné v Internetu)
  - pro privátní použití
  - může použít každý v Internetu, adresy se nepoužívají mimo Vaši síť
  - Třída A: 10.0.0.0–10.255.255.255
  - Třída B: 172.16.0.0–172.31.255.255
  - Třída C: 192.168.0.0–192.168.255.255
- Special-Use addresses
  - RFC 3330
    - Několik dalších adres sítí rezervovaných
    - 169.254.0.0/16 „link-local address block“ aka Zeroconf viz. RFC 3927

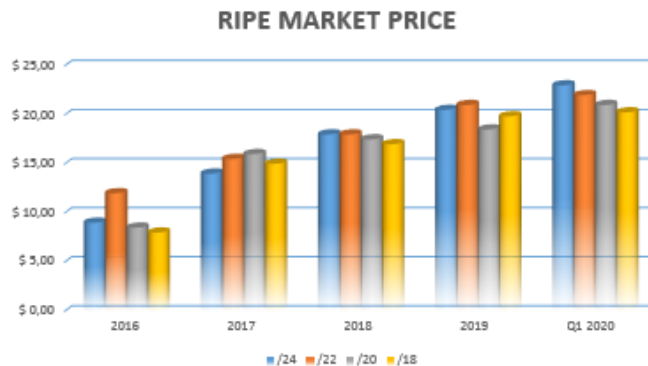
# Rezervované adresy

| Address block      | Range                         | Number of addresses | Scope           | Purpose                                                                                                                                                                                                                  |
|--------------------|-------------------------------|---------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0.0.0/8          | 0.0.0.0 – 255.255.255         | 16,777,216          | Software        | Used for broadcast messages to the current ("this") <sup>[1]</sup>                                                                                                                                                       |
| 10.0.0.0/8         | 10.0.0.0 – 10.255.255.255     | 16,777,216          | Private network | Used for local communications within a <a href="#">private network</a> <sup>[2]</sup>                                                                                                                                    |
| 100.64.0.0/10      | 100.64.0.0 – 100.127.255.255  | 4,194,304           | Private network | Used for communications between a service provider and its subscribers when using a <a href="#">carrier-grade NAT</a> <sup>[3]</sup>                                                                                     |
| 127.0.0.0/8        | 127.0.0.0 – 127.255.255.255   | 16,777,216          | Host            | Used for <a href="#">loopback addresses</a> to the local host <sup>[4]</sup>                                                                                                                                             |
| 169.254.0.0/16     | 169.254.0.0 – 169.254.255.255 | 65,536              | Subnet          | Used for <a href="#">link-local addresses</a> between two hosts on a single link when no IP address is otherwise specified, such as would have normally been retrieved from a <a href="#">DHCP</a> server <sup>[5]</sup> |
| 172.16.0.0/12      | 172.16.0.0 – 172.31.255.255   | 1,048,576           | Private network | Used for local communications within a private network <sup>[2]</sup>                                                                                                                                                    |
| 192.0.0.0/24       | 192.0.0.0 – 192.0.0.255       | 256                 | Private network | Used for the IANA IPv4 Special Purpose Address Registry <sup>[6]</sup>                                                                                                                                                   |
| 192.0.2.0/24       | 192.0.2.0 – 192.0.2.255       | 256                 | Documentation   | Assigned as "TEST-NET" for use in <a href="#">documentation</a> and examples. It should not be used publicly. <sup>[7]</sup>                                                                                             |
| 192.88.99.0/24     | 192.88.99.0 – 192.88.99.255   | 256                 | Internet        | Used by <a href="#">6to4 anycast</a> relays <sup>[8]</sup>                                                                                                                                                               |
| 192.168.0.0/16     | 192.168.0.0 – 192.168.255.255 | 65,536              | Private network | Used for local communications within a private network <sup>[2]</sup>                                                                                                                                                    |
| 198.18.0.0/15      | 198.18.0.0 – 198.19.255.255   | 131,072             | Private network | Used for testing of inter-network communications between two separate subnets <sup>[9]</sup>                                                                                                                             |
| 198.51.100.0/24    | 198.51.100.0 – 198.51.100.255 | 256                 | Documentation   | Assigned as "TEST-NET-2" for use in <a href="#">documentation</a> and examples. It should not be used publicly. <sup>[7]</sup>                                                                                           |
| 203.0.113.0/24     | 203.0.113.0 – 203.0.113.255   | 256                 | Documentation   | Assigned as "TEST-NET-3" for use in <a href="#">documentation</a> and examples. It should not be used publicly. <sup>[7]</sup>                                                                                           |
| 224.0.0.0/4        | 224.0.0.0 – 239.255.255.255   | 268,435,456         | Internet        | Reserved for multicast <sup>[10]</sup>                                                                                                                                                                                   |
| 240.0.0.0/4        | 240.0.0.0 – 255.255.255.254   | 268,435,456         | Internet        | Reserved for future use <sup>[11]</sup>                                                                                                                                                                                  |
| 255.255.255.255/32 | 255.255.255.255               | 1                   | Subnet          | Reserved for the "limited broadcast" destination address <sup>[11]</sup>                                                                                                                                                 |

# Nedostatek IPv4 adres



- Spekulování s IP adresami na prodej či pronájem je legitimní business
- <https://www.google.com/search?q=ipv4+addresses+for+lease>

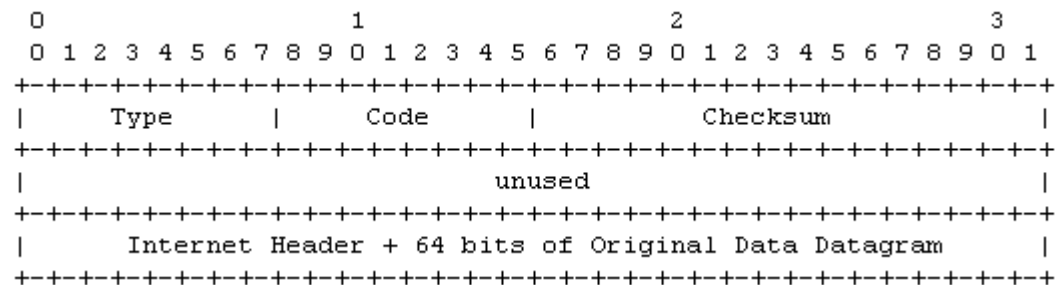


# Obsah

- Služby a funkce síťové vrstvy
- Protokol IPv4
  - Funkce, formát
  - Fragmentace a znovusestavení
  - Adresování v IPv4
- Správa adresního prostoru
- Protokol ICMP
  - Funkce, formát
  - Příklady služeb využívající ICMP
- Protokol DHCP
  - Funkce, formát
- Překlad adres, NAT

# ICMP protokol

- Internet Control Message Protocol
- Používán hosty, směrovači pro oznámení chyb, diagnostiku sítě
  - Požadovaná služba není dosažitelná
  - Cílová síť/host je nedosažitelná
- RFC 792, STD [5]
- Obecně 2 typy zpráv
  - Oznámení chyby
  - Dotaz na službu
- ICMP data přenášena v IP datagramu

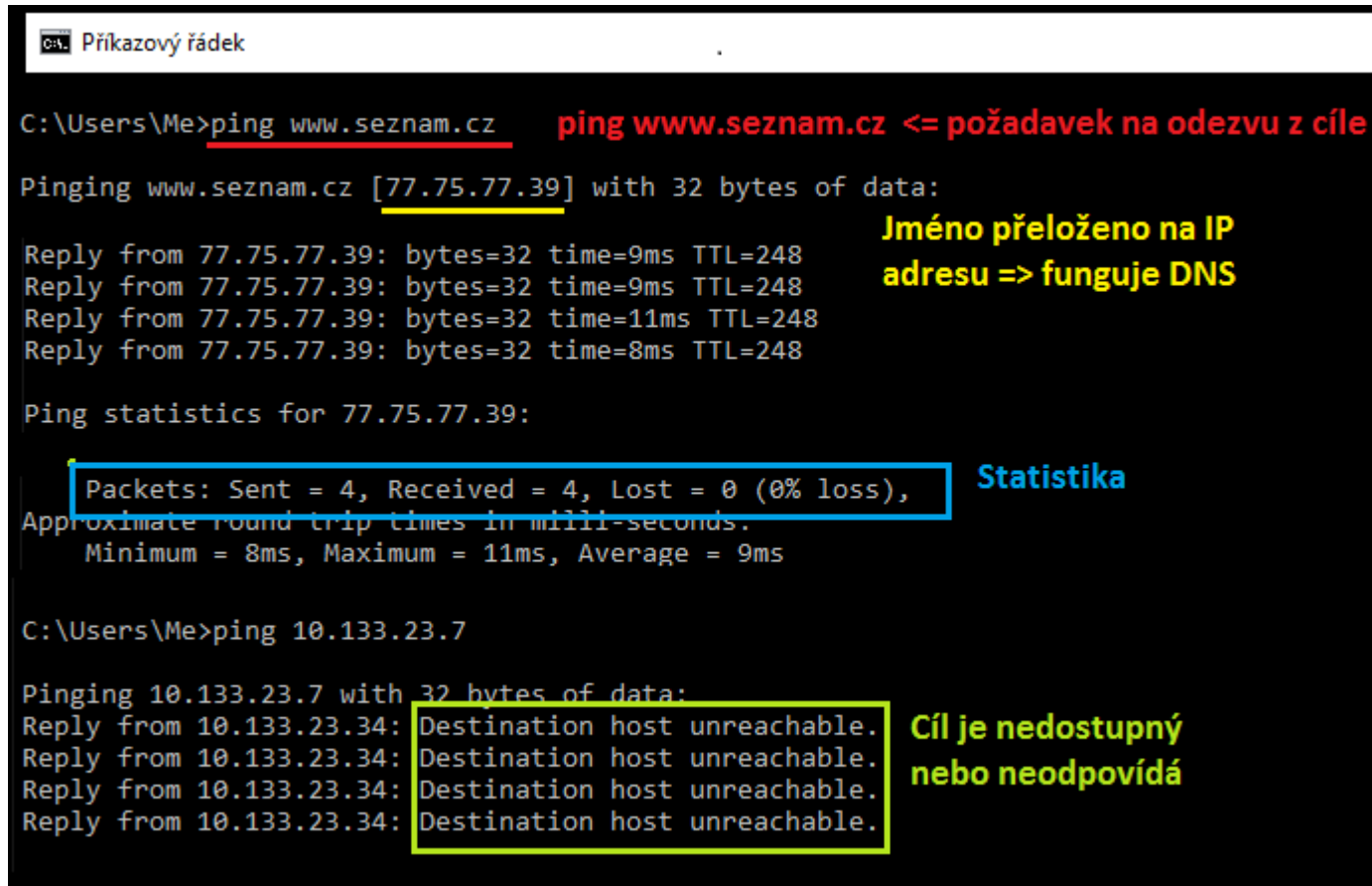


# ICMP protokol

| Type | Code | Description                                   |
|------|------|-----------------------------------------------|
| 0    | 0    | echo reply (ping)                             |
| 3    | 0    | dest. network unreachable                     |
| 3    | 1    | dest host unreachable                         |
| 3    | 2    | dest protocol unreachable                     |
| 3    | 3    | dest port unreachable                         |
| 3    | 6    | dest network unknown                          |
| 3    | 7    | dest host unknown                             |
| 4    | 0    | source quench (congestion control - not used) |
| 8    | 0    | echo request (ping)                           |
| 9    | 0    | route advertisement                           |
| 10   | 0    | router discovery                              |
| 11   | 0    | TTL expired                                   |
| 12   | 0    | bad IP header                                 |

# Ping Aplikace

- ping



The screenshot shows a Windows command prompt window titled "Příkazový řádek". The user has entered the command `C:\Users\Me>ping www.seznam.cz`. The output shows the IP address `77.75.77.39` and four successful replies with response times ranging from 8ms to 11ms. A yellow box highlights the IP address, and a yellow note states: "Jméno přeloženo na IP adresu => funguje DNS". Below the replies, the ping statistics are shown: "Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)", "Minimum = 8ms, Maximum = 11ms, Average = 9ms". A blue box highlights the statistics, and a blue note states: "Statistika". The user then enters the command `C:\Users\Me>ping 10.133.23.7`. The output shows four failed replies with the message "Destination host unreachable." A yellow box highlights these messages, and a yellow note states: "Cíl je nedostupný nebo neodpovídá".

```
C:\Users\Me>ping www.seznam.cz    ping www.seznam.cz <= požadavek na odezvu z cíle

Pinging www.seznam.cz [77.75.77.39] with 32 bytes of data:

Reply from 77.75.77.39: bytes=32 time=9ms TTL=248
Reply from 77.75.77.39: bytes=32 time=9ms TTL=248
Reply from 77.75.77.39: bytes=32 time=11ms TTL=248
Reply from 77.75.77.39: bytes=32 time=8ms TTL=248

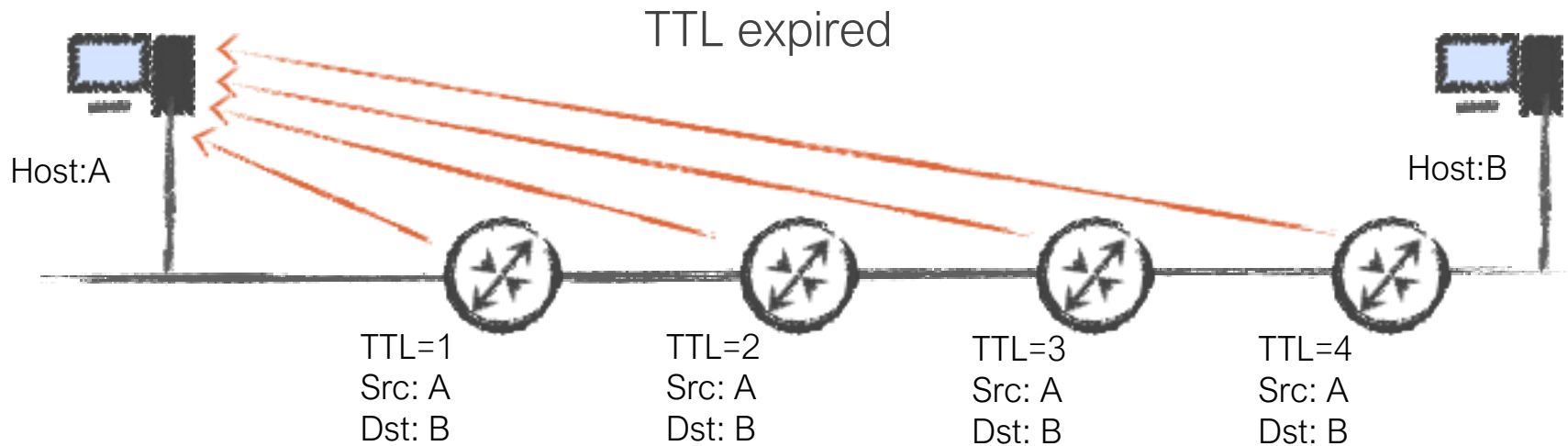
Ping statistics for 77.75.77.39:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 11ms, Average = 9ms

C:\Users\Me>ping 10.133.23.7

Pinging 10.133.23.7 with 32 bytes of data:
Reply from 10.133.23.34: Destination host unreachable.
Reply from 10.133.23.34: Destination host unreachable.
Reply from 10.133.23.34: Destination host unreachable.
Reply from 10.133.23.34: Destination host unreachable.
```

<http://www.pvfree.net/obrazky/diag-ping.png>

# Traceroute



# TraceRoute Aplikace

- Windows: `tracert`
- Unix: `traceroute`, `mtr`

```
C:\WINDOWS\system32\cmd.exe

C:\Users\Mordeth>tracert www.seznam.cz

Tracing route to www.seznam.cz [77.75.79.53]
over a maximum of 30 hops:

  1  <1 ms    <1 ms    <1 ms    10.20.40.1
  2   1 ms    <1 ms    <1 ms    gw.vcit.vutbr.net [185.62.108.65]
  3   1 ms     1 ms    <1 ms    pe-ant.net.vutbr.cz [147.229.254.154]
  4  <1 ms    <1 ms    <1 ms    rt-ant.net.vutbr.cz [147.229.253.233]
  5   3 ms     3 ms     3 ms    213.195.192.247
  6   4 ms     3 ms     4 ms    nix4.seznam.cz [91.210.16.195]
  7   4 ms     4 ms     4 ms    n7k-ko-a-vdc-1-po1.seznam.cz [185.66.188.7]
  8   4 ms     4 ms     4 ms    n7k-ko-a-vdc-2-po3.seznam.cz [185.66.188.25]
  9   4 ms     3 ms     4 ms    www.seznam.cz [77.75.79.53]

Trace complete.
```

|                          |     |      |      |       |       |      |    |      |      |      |           |
|--------------------------|-----|------|------|-------|-------|------|----|------|------|------|-----------|
| Tue Oct 8 14:11:16 2013  |     |      |      |       |       |      |    |      |      |      |           |
| Pings                    |     |      |      |       |       |      |    |      |      |      |           |
| Host                     | Avg | Best | Wrst | StDev |       |      |    |      |      |      |           |
| 9. 38.104.103.254        | 0.5 | 0.5  | 0.3  | 0.7   | 0.1   |      |    |      |      |      |           |
| 38.104.103.130           | 0.8 | 1.5  | 0.6  | 24.9  | 3.9   |      |    |      |      |      |           |
| 10. 209.85.254.128       | 8.6 | 9.4  | 3.3  | 120.5 | 24.5  |      |    |      |      |      |           |
| 209.85.250.30            | 8.8 | 80.3 | 3.5  | 555.2 | 146.4 |      |    |      |      |      |           |
| ord08s07-in-f1.1e100.net | 8.7 | 30.0 | 3.5  | 204.0 | 57.1  |      |    |      |      |      |           |
|                          | 6.2 | 27.9 | 4.9  | 199.9 | 49.8  |      |    |      |      |      |           |
|                          | 1.1 | 11.3 | 10.9 | 12.5  | 0.4   |      |    |      |      |      |           |
|                          | 1.4 | 11.7 | 11.2 | 13.2  | 0.4   |      |    |      |      |      |           |
|                          |     |      |      |       |       | 0.0% | 38 | 17.6 | 17.8 | 10.1 | 82.0 15.5 |
|                          |     |      |      |       |       | 5.3% | 38 | 10.2 | 12.2 | 10.1 | 26.8 4.0  |
|                          |     |      |      |       |       | 0.0% | 38 | 10.9 | 10.9 | 10.4 | 15.1 0.7  |
|                          |     |      |      |       |       | 0.0% | 38 | 10.2 | 10.5 | 10.2 | 13.8 0.6  |

# ICMP ve Wiresharku

The image shows a Wireshark network traffic capture of ICMP ping requests and replies. The packet list pane at the top shows 19 packets, all of which are ICMP Echo (ping) requests or replies. The packet details pane on the left shows the structure of the selected packet (Frame 50), including the Ethernet II header, Internet Protocol Version 4 header, and Internet Control Message Protocol (ICMP) header. The packet bytes pane at the bottom shows the raw data of the selected packet, including the ICMP header and the data field.

Filter: icmp

| No. | Time            | Source       | Destination  | Protocol | Length | Info                                                 |
|-----|-----------------|--------------|--------------|----------|--------|------------------------------------------------------|
| 50  | 13:00:56.200112 | 192.168.0.18 | 143.89.14.34 | ICMP     | 74     | Echo (ping) request id=0x0001, seq=69/17664, ttl=128 |
| 51  | 13:00:56.437860 | 143.89.14.34 | 192.168.0.18 | ICMP     | 74     | Echo (ping) reply id=0x0001, seq=69/17664, ttl=234   |
| 61  | 13:00:57.210700 | 192.168.0.18 | 143.89.14.34 | ICMP     | 74     | Echo (ping) request id=0x0001, seq=70/17920, ttl=128 |
| 97  | 13:00:57.445707 | 143.89.14.34 | 192.168.0.18 | ICMP     | 74     | Echo (ping) reply id=0x0001, seq=70/17920, ttl=234   |
| 100 | 13:00:58.224522 | 192.168.0.18 | 143.89.14.34 | ICMP     | 74     | Echo (ping) request id=0x0001, seq=71/18176, ttl=128 |
| 101 | 13:00:58.464578 | 143.89.14.34 | 192.168.0.18 | ICMP     | 74     | Echo (ping) reply id=0x0001, seq=71/18176, ttl=234   |
| 107 | 13:00:59.222809 | 192.168.0.18 | 143.89.14.34 | ICMP     | 74     | Echo (ping) request id=0x0001, seq=72/18432, ttl=128 |
| 108 | 13:00:59.459167 | 143.89.14.34 | 192.168.0.18 | ICMP     | 74     | Echo (ping) reply id=0x0001, seq=72/18432, ttl=234   |
| 109 | 13:01:00.221311 | 192.168.0.18 | 143.89.14.34 | ICMP     | 74     | Echo (ping) request id=0x0001, seq=73/18688, ttl=128 |
| 150 | 13:01:00.456626 | 143.89.14.34 | 192.168.0.18 | ICMP     | 74     | Echo (ping) reply id=0x0001, seq=73/18688, ttl=234   |
| 158 | 13:01:01.251312 | 192.168.0.18 | 143.89.14.34 | ICMP     | 74     | Echo (ping) request id=0x0001, seq=74/18944, ttl=128 |
| 159 | 13:01:01.493802 | 143.89.14.34 | 192.168.0.18 | ICMP     | 74     | Echo (ping) reply id=0x0001, seq=74/18944, ttl=234   |
| 183 | 13:01:02.255017 | 192.168.0.18 | 143.89.14.34 | ICMP     | 74     | Echo (ping) request id=0x0001, seq=75/19200, ttl=128 |
| 184 | 13:01:02.490204 | 143.89.14.34 | 192.168.0.18 | ICMP     | 74     | Echo (ping) reply id=0x0001, seq=75/19200, ttl=234   |
| 190 | 13:01:03.263590 | 192.168.0.18 | 143.89.14.34 | ICMP     | 74     | Echo (ping) request id=0x0001, seq=76/19456, ttl=128 |

Frame 50: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0  
Ethernet II, Src: IntelCor\_a2:16:ed (00:21:5c:a2:16:ed), Dst: Netgear\_77:1a:70 (00:0f:b5:77:1a:70)  
Internet Protocol Version 4, Src: 192.168.0.18 (192.168.0.18), Dst: 143.89.14.34 (143.89.14.34)  
Version: 4  
Header length: 20 bytes  
Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))  
Total Length: 60  
Identification: 0x1264 (4708)  
Flags: 0x00  
Fragment offset: 0  
Time to live: 128  
Protocol: ICMP (1)  
Header checksum: 0xca27 [correct]  
Source: 192.168.0.18 (192.168.0.18)  
Destination: 143.89.14.34 (143.89.14.34)  
Internet Control Message Protocol  
Type: 8 (Echo (ping) request)  
Code: 0  
Checksum: 0x4d16 [correct]  
Identifier (BE): 1 (0x0001)  
Identifier (LE): 256 (0x0100)  
Sequence number (BE): 69 (0x0045)  
Sequence number (LE): 17664 (0x4500)

Command Prompt

```
C:\Users\Zack>
C:\Users\Zack>
C:\Users\Zack>
C:\Users\Zack>ping -n 10 www.ust.hk

Pinging www.ust.hk [143.89.14.34] with 32 bytes of data:
Reply from 143.89.14.34: bytes=32 time=238ms TTL=234
Reply from 143.89.14.34: bytes=32 time=235ms TTL=234
Reply from 143.89.14.34: bytes=32 time=240ms TTL=234
Reply from 143.89.14.34: bytes=32 time=236ms TTL=234
Reply from 143.89.14.34: bytes=32 time=235ms TTL=234
Reply from 143.89.14.34: bytes=32 time=242ms TTL=234
Reply from 143.89.14.34: bytes=32 time=235ms TTL=234
Reply from 143.89.14.34: bytes=32 time=235ms TTL=234
Reply from 143.89.14.34: bytes=32 time=238ms TTL=234
Reply from 143.89.14.34: bytes=32 time=235ms TTL=234

Ping statistics for 143.89.14.34:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 235ms, Maximum = 242ms, Average = 236ms

C:\Users\Zack>
C:\Users\Zack>
C:\Users\Zack>
```

Checksum (icmp.checksum), 2 bytes

Packets: 301 Displayed: 20 Marked: 0 Dropped: 0

Profile: Default

<https://zaccouellet.files.wordpress.com/2012/05/wsicmp.png>

# Obsah

- Služby a funkce síťové vrstvy
- Protokol IPv4
  - Funkce, formát
  - Fragmentace a znovusestavení
  - Adresování v IPv4
- Správa adresního prostoru
- Protokol ICMP
  - Funkce, formát
  - Příklady služeb využívající ICMP
- Protokol DHCP
  - Funkce, formát
- Překlad adres, NAT

# DHCP

- Dynamic Host Configuration Protocol
- Rozšíření původního BOOTP protokolu (pro bezdiskové stanice)
- DHCP klient / DHCP server
- Poskytuje:
  - Mechanismus přidělení IP adresy a TCP/IP parametrů
  - Dohodu a přenos specifických informací pro hosta
- Aplikační protokol !
- UDP, porty 67 a 68

# DHCP Discover

## DHCP Discover

### ETH:

src mac: AA:AA:AA:AA:AA:AA  
dst mac: FF:FF:FF:FF:FF:FF  
(broadcast)

### IP

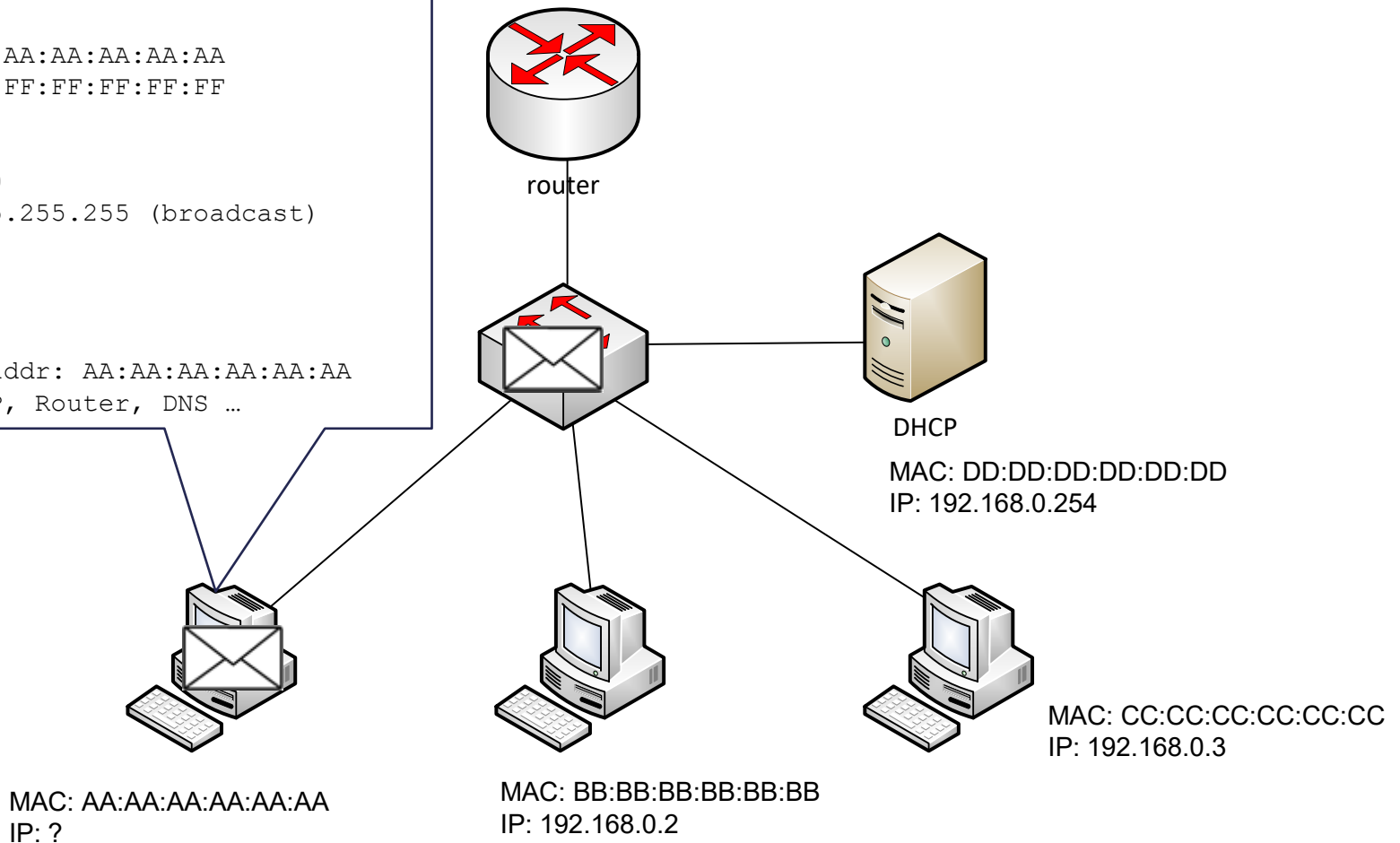
src: 0.0.0.0  
dst: 255.255.255.255 (broadcast)

### UDP

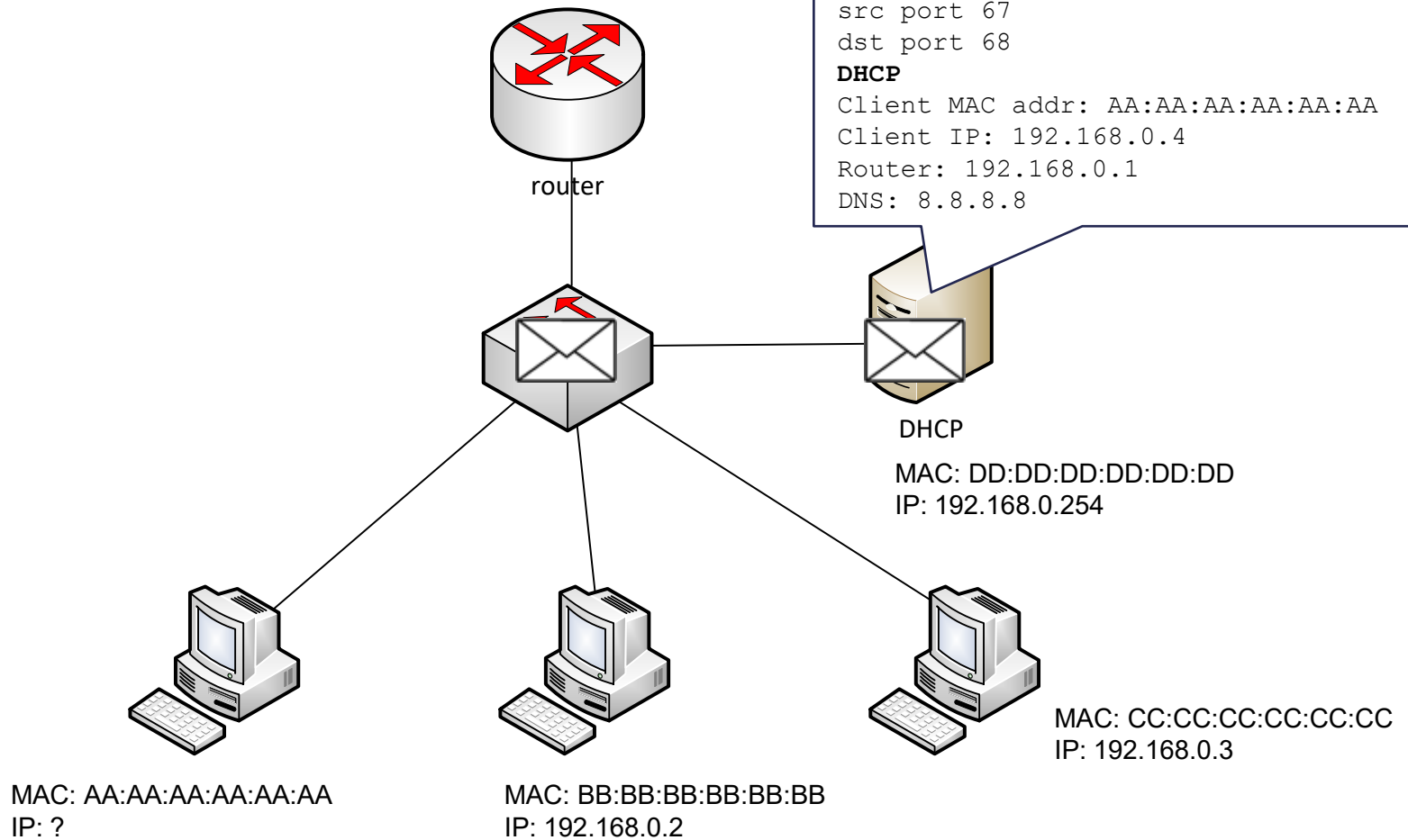
src port 68  
dst port 67

### DHCP

Client MAC addr: AA:AA:AA:AA:AA:AA  
Requests: IP, Router, DNS ...



# DHCP Offer



# DHCP Request

## DHCP Request

### ETH:

src mac: AA:AA:AA:AA:AA:AA

dst mac: FF:FF:FF:FF:FF:FF

(broadcast)

### IP

src: 0.0.0.0

dst: 255.255.255.255 (broadcast)

### UDP

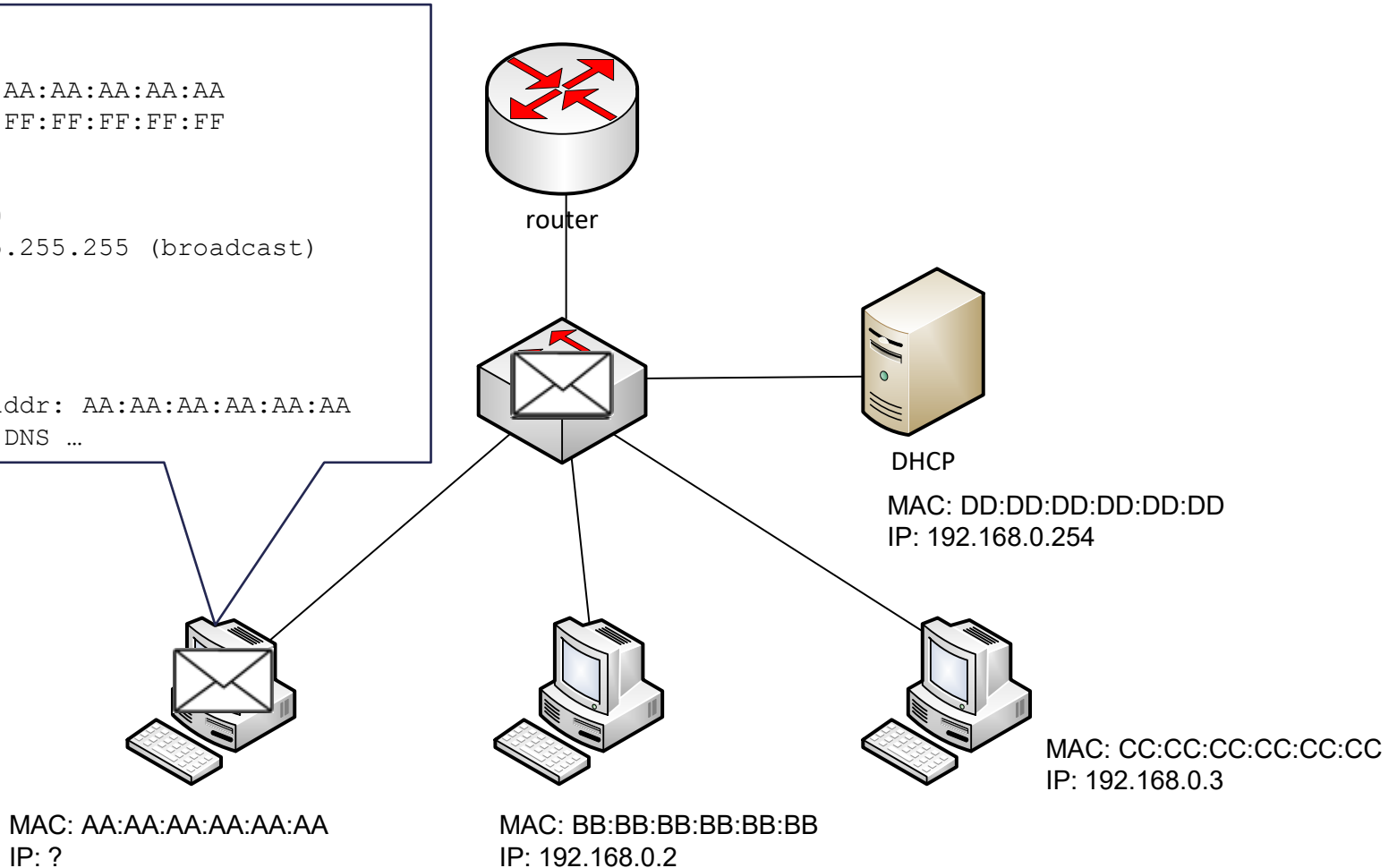
src port 68

dst port 67

### DHCP

Client MAC addr: AA:AA:AA:AA:AA:AA

IP, Router, DNS ...



# DHCP Ack

## DHCP Acknowledgement

### ETH:

src mac: DD:DD:DD:DD:DD:DD

dst mac: AA:AA:AA:AA:AA:AA

### IP

src: 192.168.0.254

dst: 192.168.0.4

### UDP

src port 67

dst port 68

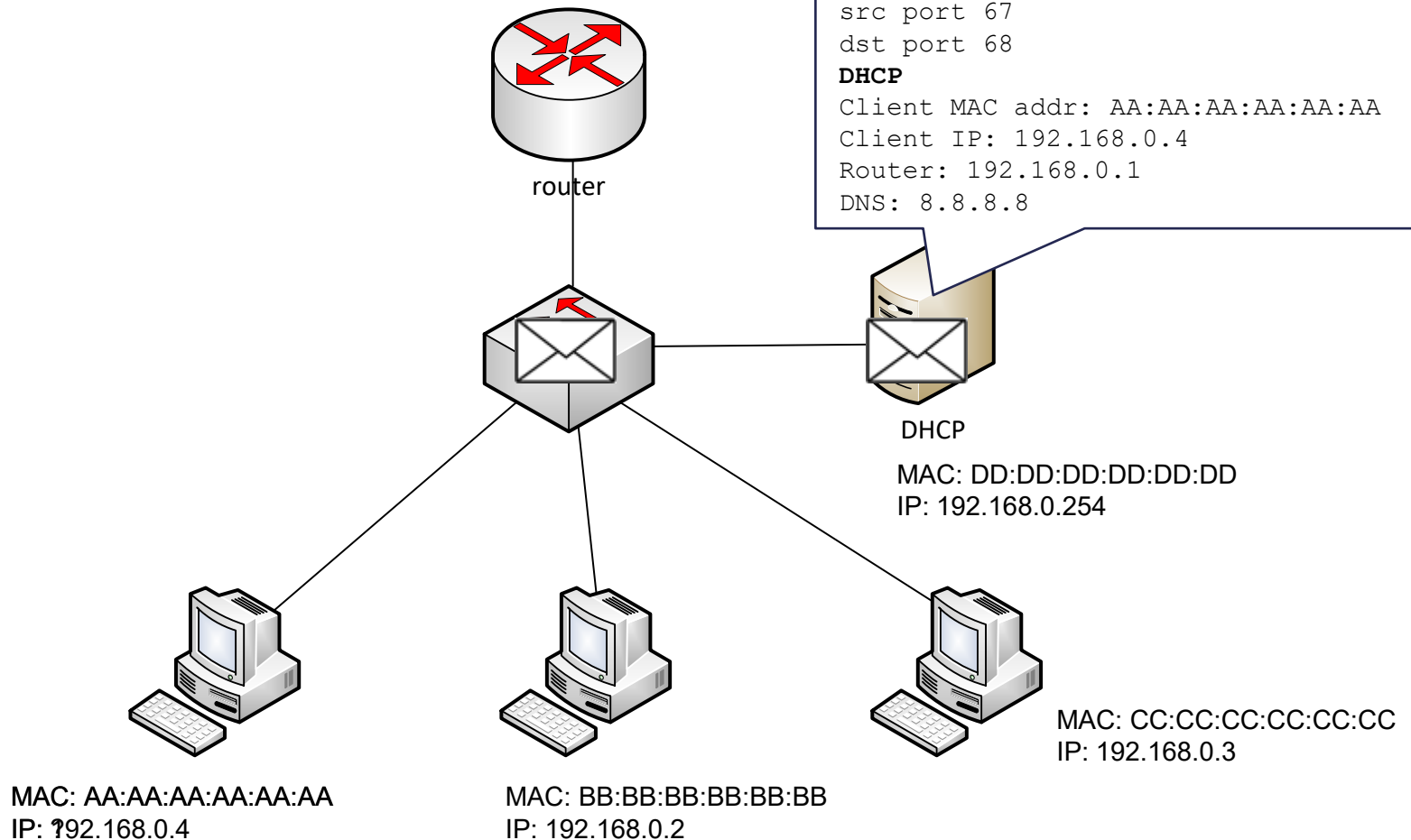
### DHCP

Client MAC addr: AA:AA:AA:AA:AA:AA

Client IP: 192.168.0.4

Router: 192.168.0.1

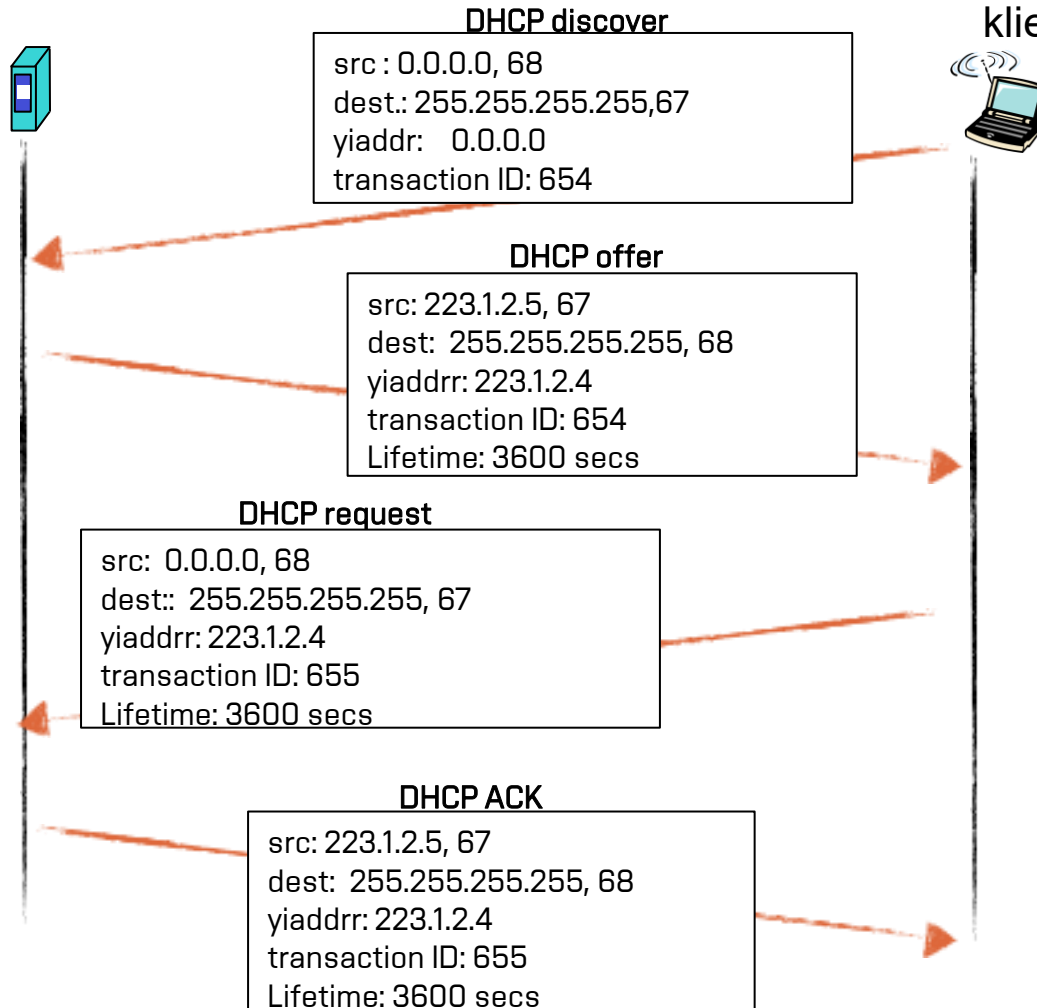
DNS: 8.8.8.8



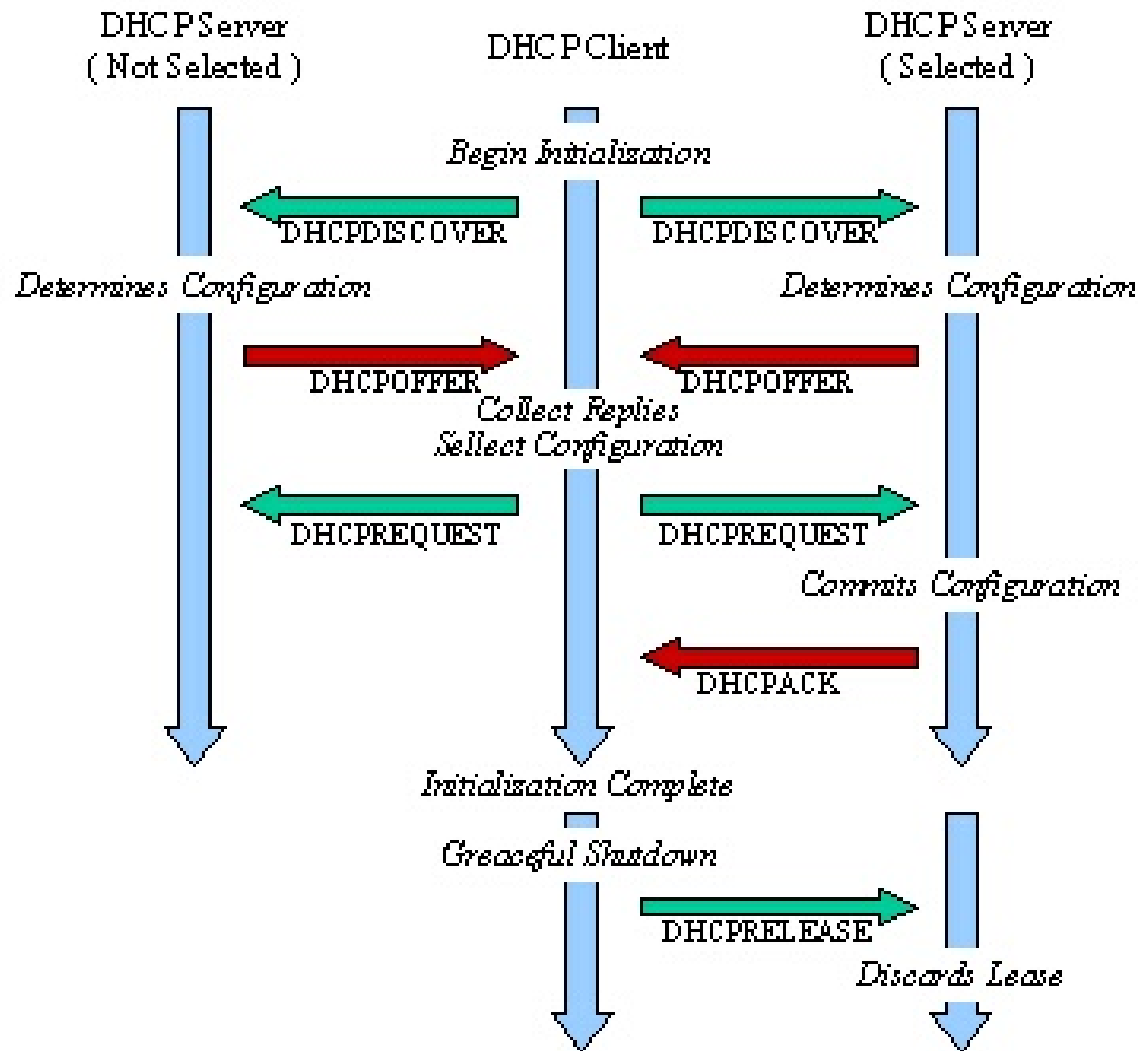
# DHCP Server Bcast příklad

DHCP server: 223.1.2.5

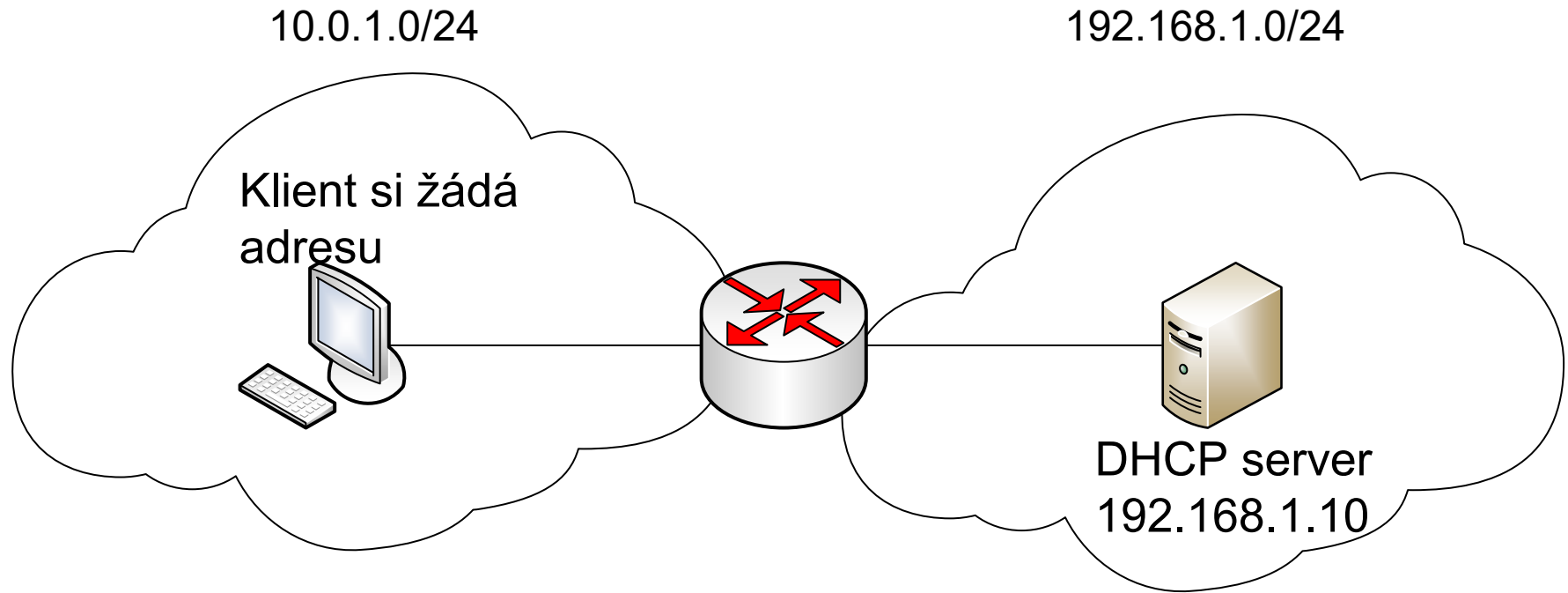
příchozí klient



# Proč DHCP broadcastuje?

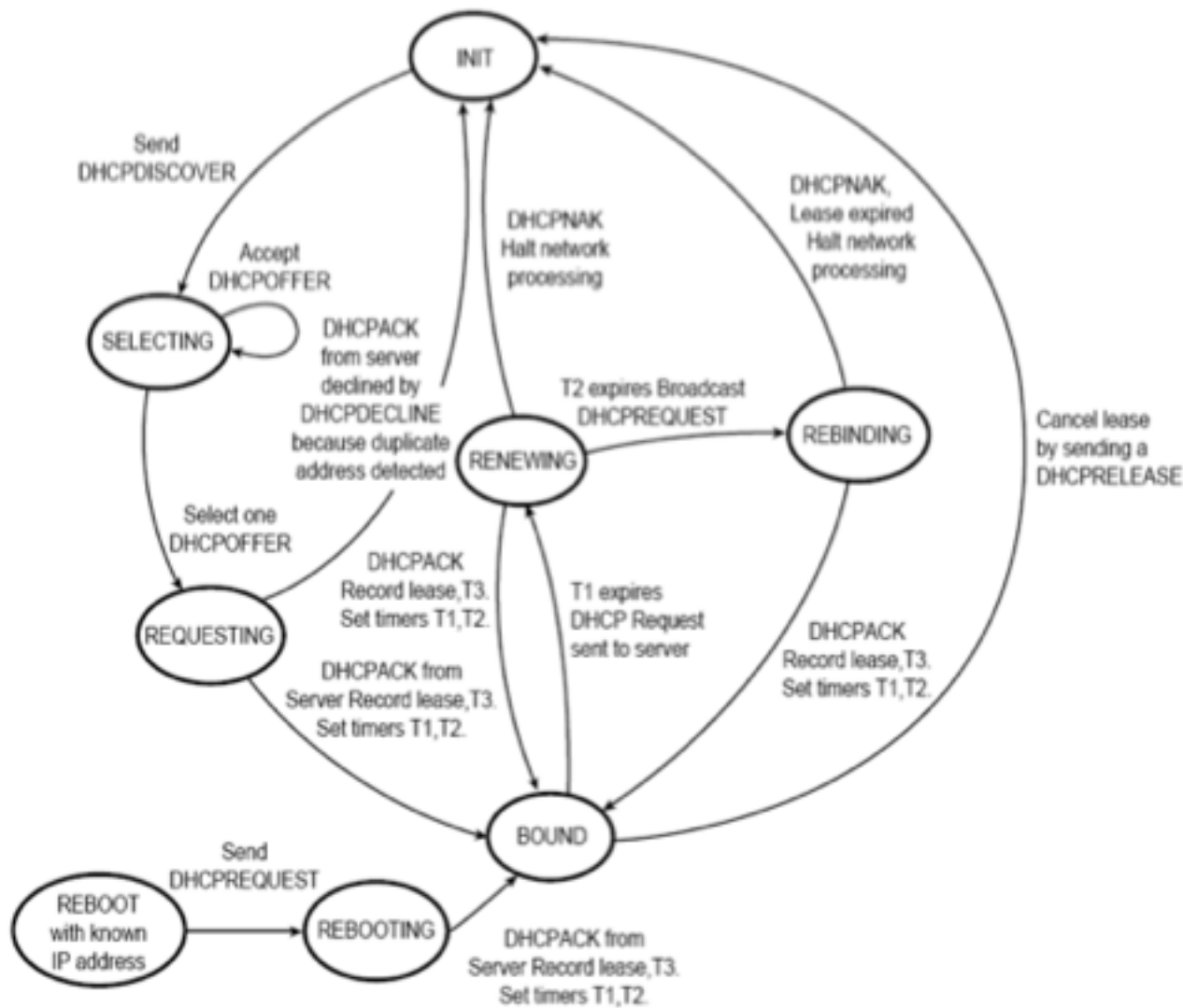


# DHCP mimo subnet



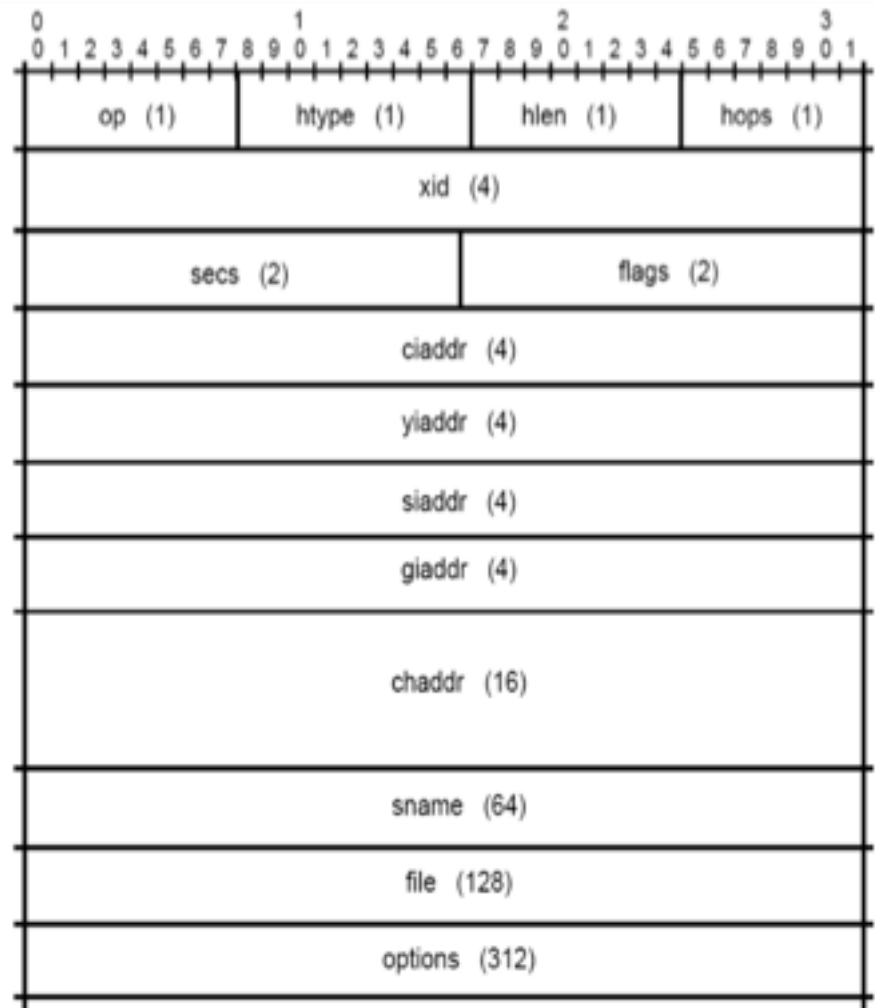
- Option-82

# DHCP stavy a interakce



# Formát DHCP zprávy

- op: dotaz nebo odpověď
- htype: typ hw adresy
- hlen: délka hw adresy v oktetech
- hops: počet skoku, pro DHCP relay
- xid: Transaction ID, náhodné číslo transakce
- secs: počet sekund od počátku bootování
- flags: broadcast nebo unicast
- ciaadr: IP adresa klienta (v odpovědi)
- yiaddr: Your IP address, přiděluje server klientovi
- siaddr: IP adresa serveru
- giaddr: IP adresa relay agenta
- chaddr: HW adresa klienta
- sname: volitelně jméno serveru
- file: image filename
- options: další volby



Numbers in parentheses ( ) represent octets

# DHCP ve Wiresharku

File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help

Filter: **bootp** Expression... Clear Apply

| No. | Time      | Source        | Destination     | Protocol | Length | Info                                      |
|-----|-----------|---------------|-----------------|----------|--------|-------------------------------------------|
| 2   | 7.587185  | 0.0.0.0       | 255.255.255.255 | DHCP     | 342    | DHCP Discover - Transaction ID 0x3e50ce3  |
| 4   | 8.632950  | 192.168.1.1   | 255.255.255.255 | DHCP     | 590    | DHCP Offer - Transaction ID 0x3e50ce3     |
| 5   | 8.633123  | 0.0.0.0       | 255.255.255.255 | DHCP     | 342    | DHCP Request - Transaction ID 0x3e50ce3   |
| 6   | 8.635133  | 192.168.1.1   | 255.255.255.255 | DHCP     | 590    | DHCP ACK - Transaction ID 0x3e50ce3       |
| 36  | 20.134178 | 192.168.1.101 | 192.168.1.1     | DHCP     | 342    | DHCP Request - Transaction ID 0x257e55a3  |
| 37  | 20.135930 | 192.168.1.1   | 255.255.255.255 | DHCP     | 590    | DHCP ACK - Transaction ID 0x257e55a3      |
| 41  | 25.073867 | 192.168.1.101 | 192.168.1.1     | DHCP     | 342    | DHCP Release - Transaction ID 0xb7a32733  |
| 42  | 30.869153 | 0.0.0.0       | 255.255.255.255 | DHCP     | 342    | DHCP Discover - Transaction ID 0x3a5df7d9 |
| 44  | 31.908133 | 192.168.1.1   | 255.255.255.255 | DHCP     | 590    | DHCP Offer - Transaction ID 0x3a5df7d9    |
| 45  | 31.908304 | 0.0.0.0       | 255.255.255.255 | DHCP     | 342    | DHCP Request - Transaction ID 0x3a5df7d9  |
| 46  | 31.910313 | 192.168.1.1   | 255.255.255.255 | DHCP     | 590    | DHCP ACK - Transaction ID 0x3a5df7d9      |

Source port: bootps (67)  
Destination port: bootpc (68)  
Length: 556  
Checksum: 0x7223 [validation disabled]

Bootstrap Protocol

Message type: Boot Reply (2)  
Hardware type: Ethernet  
Hardware address length: 6  
Hops: 0  
Transaction ID: 0x3e50ce3  
Seconds elapsed: 0  
Bootp flags: 0x0000 (unicast)  
Client IP address: 0.0.0.0 (0.0.0.0)  
Your (client) IP address: 192.168.1.101 (192.168.1.101)  
Next server IP address: 0.0.0.0 (0.0.0.0)  
Relay agent IP address: 0.0.0.0 (0.0.0.0)  
Client MAC address: DellComp\_4f:36:23 (00:08:74:4f:36:23)  
Client hardware address padding: 00000000000000000000  
Server host name not given  
Boot file name not given  
Magic cookie: DHCP

Option: (t=53,l=1) DHCP Message Type = DHCP Offer  
Option: (53) DHCP Message Type  
Length: 1  
Value: 02

Option: (t=1,l=4) Subnet Mask = 255.255.255.0  
Option: (1) Subnet Mask  
Length: 4  
Value: ffffffff00

Option: (t=3,l=4) Router = 192.168.1.1  
Option: (3) Router  
Length: 4  
Value: c0a80101

Option: (t=6,l=8) Domain Name Server  
Option: (6) Domain Name Server  
Length: 8

0030 0c e3 00 00 00 00 00 00 00 00 00 c0 a8 01 65 00 00 .....TO6#...  
0040 00 00 00 00 00 00 00 00 00 08 74 4f 36 23 00 00 .....  
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

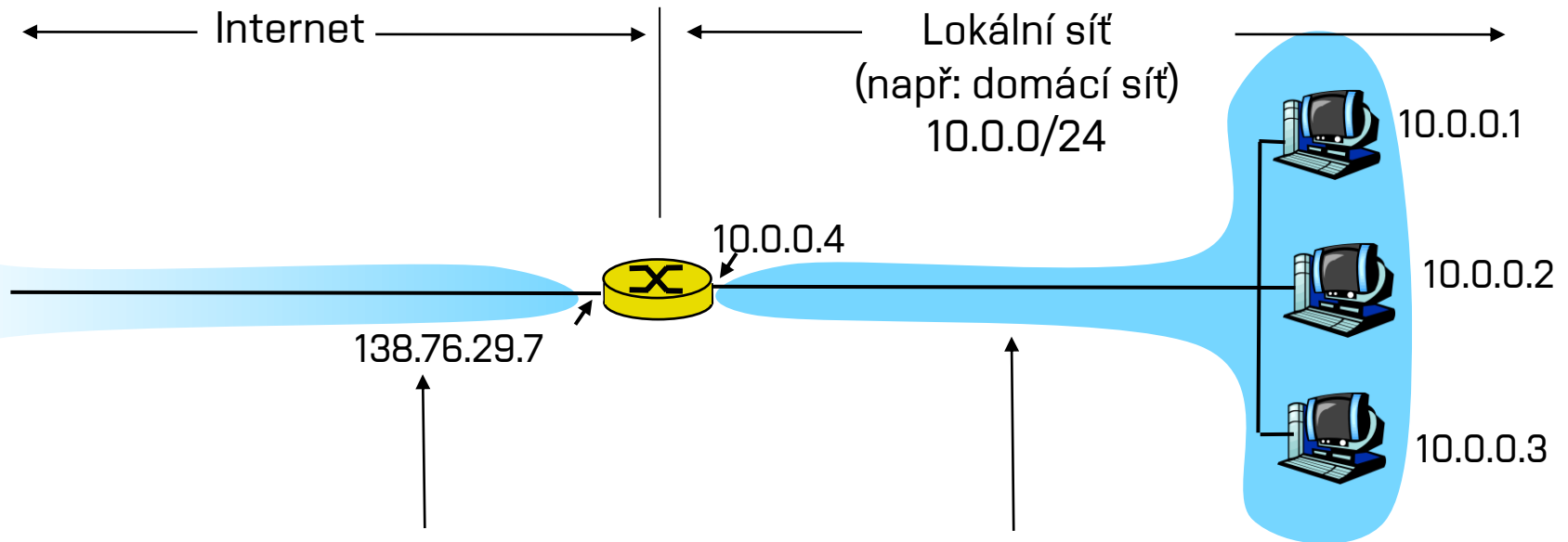
d:\... \bootp.in.vourl.4.bv... Packets: 63 Displayed: 11 Marked: 0 Load time: 0:00:062

# Obsah

- Služby a funkce síťové vrstvy
- Protokol IPv4
  - Funkce, formát
  - Fragmentace a znovusestavení
  - Adresování v IPv4
- Správa adresního prostoru
- Protokol ICMP
  - Funkce, formát
  - Příklady služeb využívající ICMP
- Protokol DHCP
  - Funkce, formát
- Překlad adres, NAT

# Překlad adres

- Network address translation, RFC 1918



*Všechny* datagramy opouští lokální síť se stejnou jednou zdrojovou NAT IP adresou: 138.76.29.7, různé zdrojové porty pro TCP/UDP

Datagramy mají zdrojovou nebo cílovou adresu v síti v rozsahu 10.0.0/24

# Zhodnocení

- **Výhody**

- Není třeba rozsah adres od poskytovatele přípojení: stačí jedna adresa pro všechna zařízení
- Můžeme měnit adresy v lokální síti aniž bychom to museli někomu sdělit/žádat
- Můžeme změnit poskytovatele (veřejnou adresu) bez nutnosti něco měnit v lokální síti

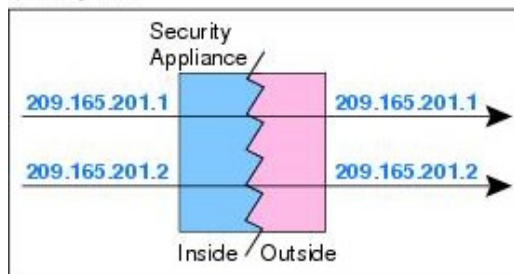
- **Nevýhody**

- Zařízení v lokální síti nejsou explicitně adresovatelné, nejsou viditelné z Internetu
- ~ bezpečnostní mechanismus (!)

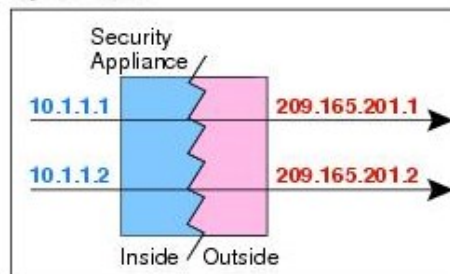
# Typy NATu

- Static NAT
  - 1:1 address mapping
- Dynamic NAT
  - N:M address mapping
  - Private address are mapped onto pool of public addresses
  - IF there is no address in public pool THEN NAT cannot work
- Overloading NAT a.k.a. Port Address Translation (PAT)
  - Dynamic NAT using one public address
  - Rewriting port numbers of TCP and UDP

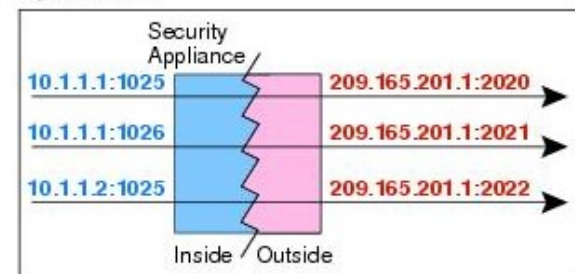
Identity NAT



Dynamic NAT



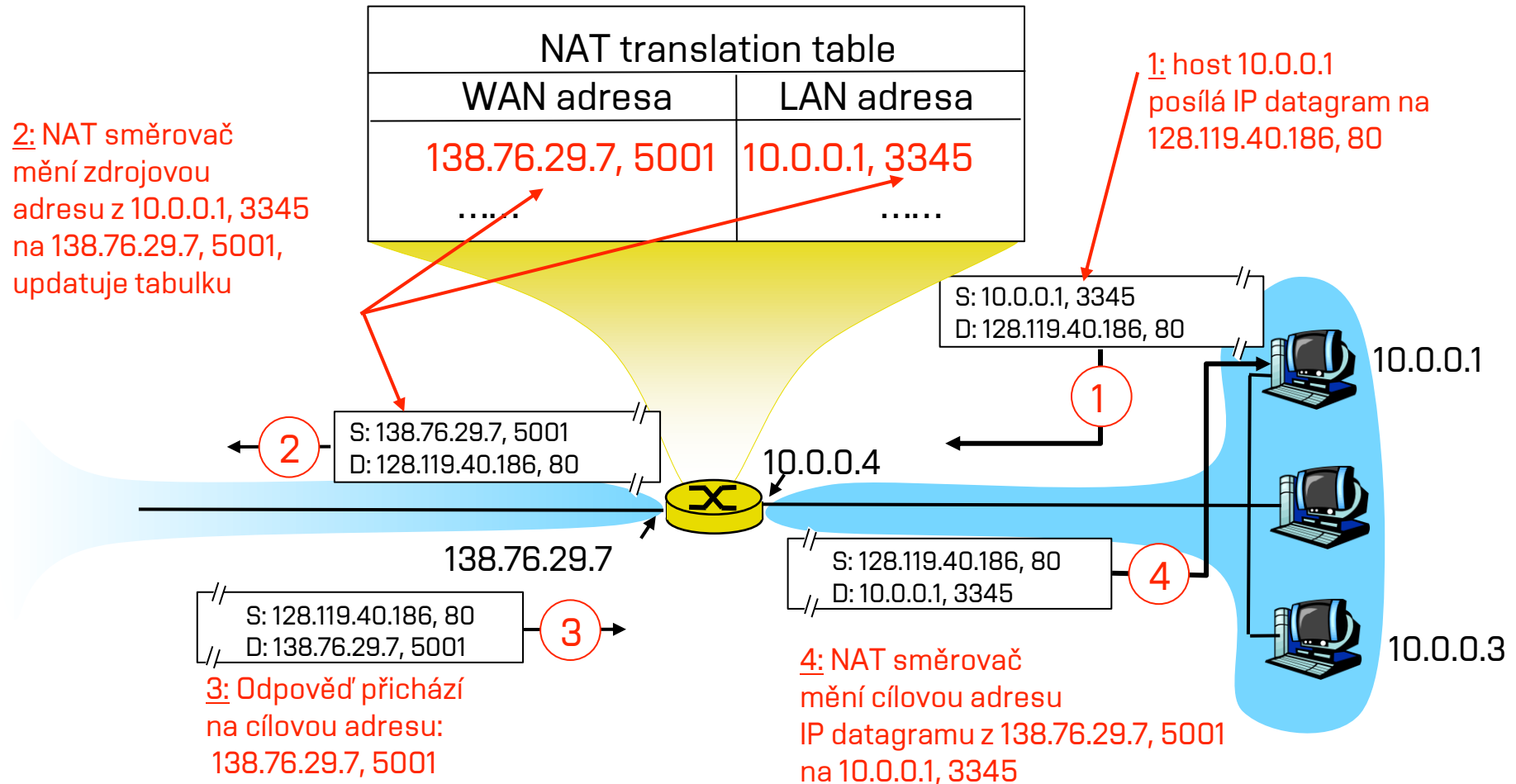
Dynamic PAT



# Překlad portů

- 16-bitové číslo portu TCP nebo UDP
  - > 65 535 současných spojení
- NAT modifikuje hlavičku
  - Pokud v aplikačním protokolu klient identifikuje svoji adresu, nastává problém (adresy se liší)
  - Aplikační protokol by měl být navržen tak, aby jej NAT neomezil
  - Router by měl pracovat na 3 vrstvě (překlady dělá obecně brána ...)

# Překládová tabulka

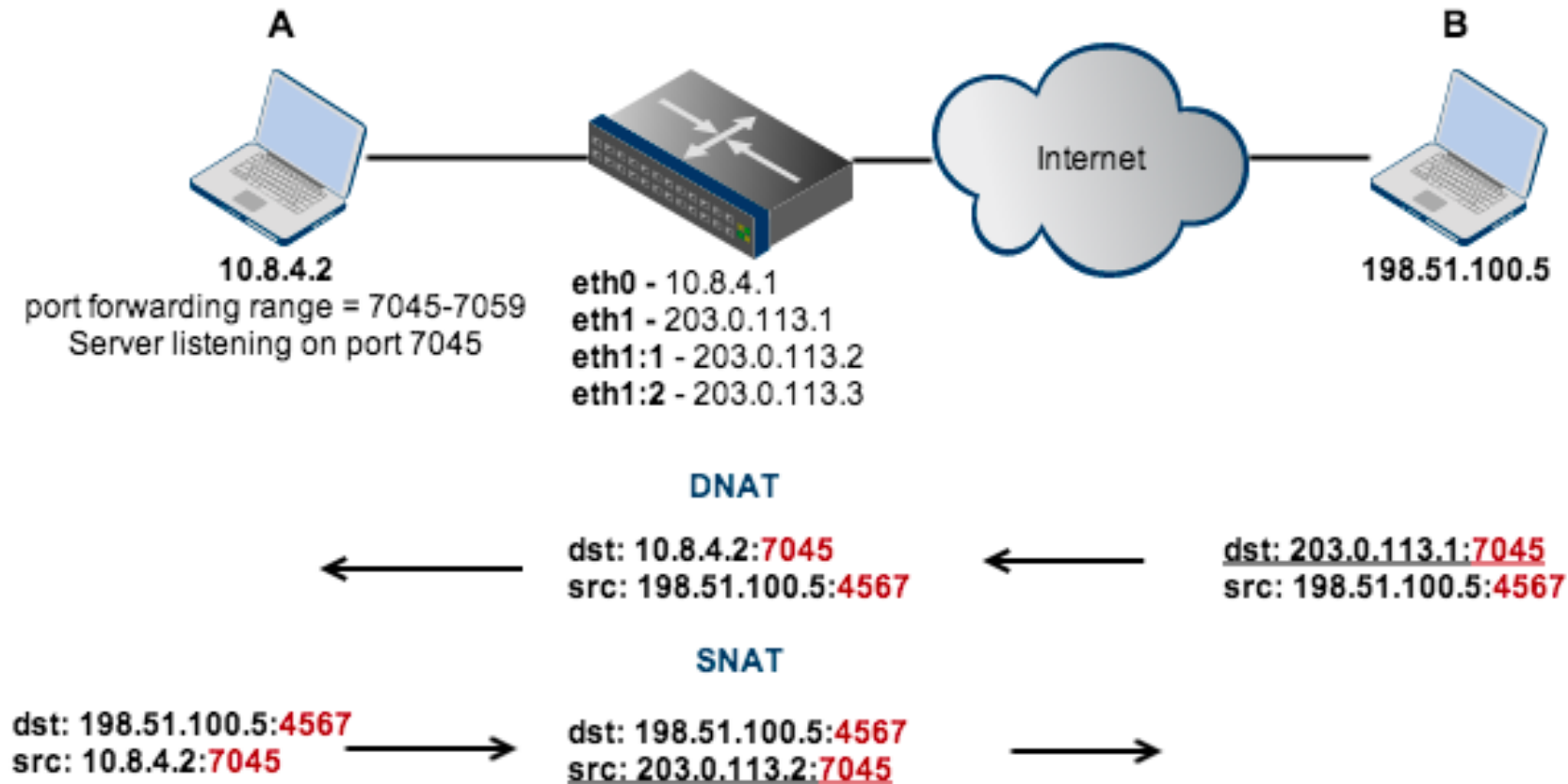


# NAT ve Wiresharku

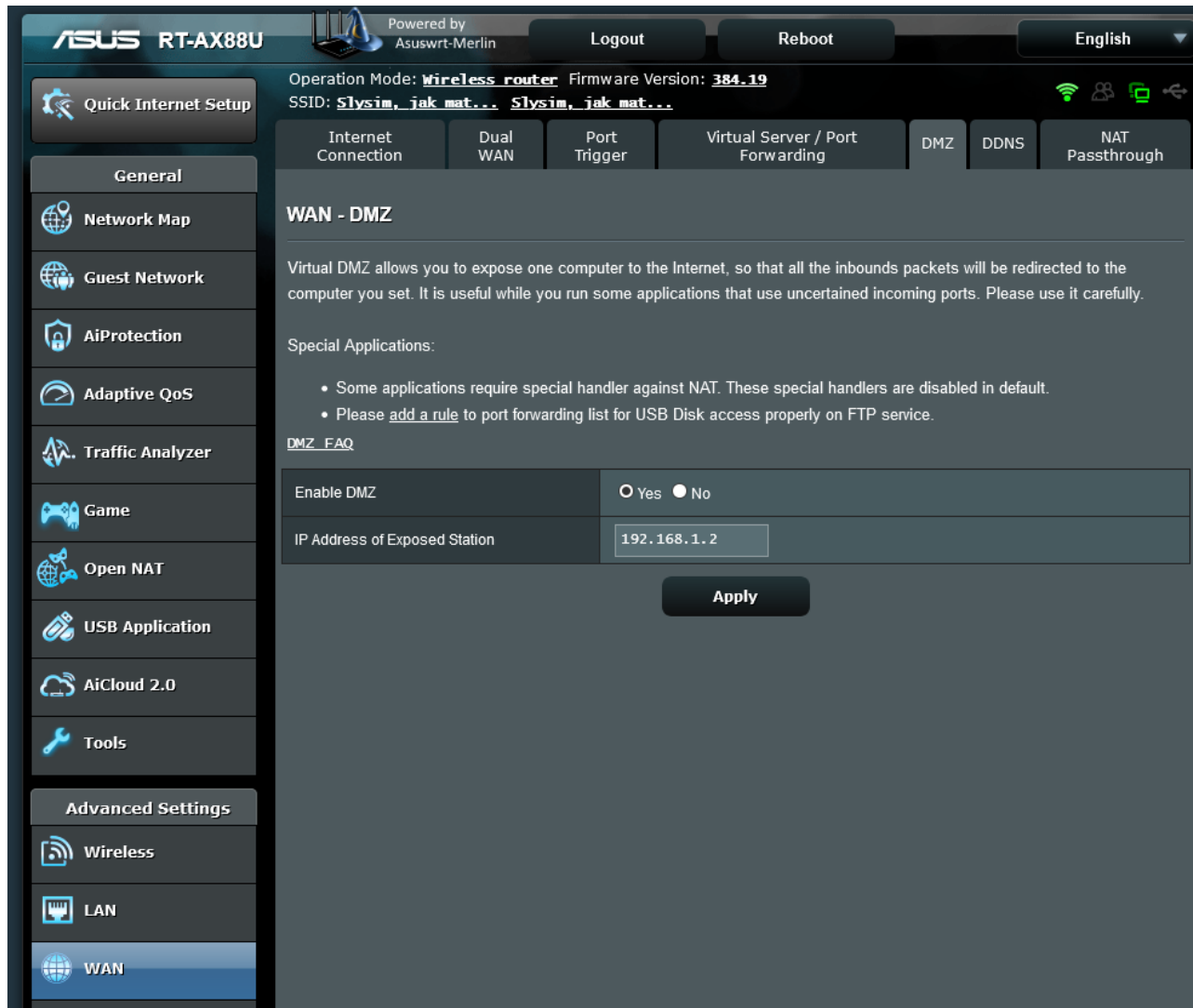
```
11 13.581074000 192.168.0.12 1.23.28.43 TCP 74 53044→22 [SYN] Seq=173468...  
Frame 11: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface  
Ethernet II, Src: LogicMod_51:66:00 (00:00:ab:51:66:00), Dst: ca:01:19:34:00:1c  
Internet Protocol Version 4, Src: 192.168.0.12 (192.168.0.12), Dst: 1.23.28.43  
Version: 4  
Header Length: 20 bytes  
Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT)  
Total Length: 60  
Identification: 0x875b (34651)  
Flags: 0x02 (Don't Fragment)  
Fragment offset: 0  
Time to live: 64  
Protocol: TCP (6)  
Header checksum: 0xd56a [validation disabled]  
Source: 192.168.0.12 (192.168.0.12)  
Destination: 1.23.28.43 (1.23.28.43)  
[Source GeoIP: Unknown]  
[Destination GeoIP: Unknown]  
Transmission Control Protocol, Src Port: 53044  
Source Port: 53044 (53044)  
Destination Port: 22 (22)  
[Stream index: 0]  
[TCP segment Len: 0]  
Sequence number: 1734683627  
Acknowledgment number: 0  
Header Length: 40 bytes  
... 0000 0000 0010 = Flags: 0x002 (SYN)  
window size value: 5840  
[Calculated window size: 5840]  
Checksum: 0x43c6 [validation disabled]  
Urgent pointer: 0  
Options: (20 bytes), Maximum segment size, SA
```

```
3 4.065077000 137.186.57.12 1.23.28.43 TCP 74 53044→22 [SYN] Seq=173468362...  
Frame 3: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface  
Ethernet II, Src: ca:01:19:34:00:08 (ca:01:19:34:00:08), Dst: ca:02:11:b4:00:08  
Internet Protocol Version 4, Src: 137.186.57.12 (137.186.57.12), Dst: 1.23.28.43  
Version: 4  
Header Length: 20 bytes  
Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT)  
Total Length: 60  
Identification: 0x875b (34651)  
Flags: 0x02 (Don't Fragment)  
Fragment offset: 0  
Time to live: 63  
Protocol: TCP (6)  
Header checksum: 0xd458 [validation disabled]  
Source: 137.186.57.12 (137.186.57.12)  
Destination: 1.23.28.43 (1.23.28.43)  
[Source GeoIP: Unknown]  
[Destination GeoIP: Unknown]  
Transmission Control Protocol, Src Port: 53044 (53044), Dst Port: 22 (22), Seq: 1734683627  
Source Port: 53044 (53044)  
Destination Port: 22 (22)  
[Stream index: 0]  
[TCP segment Len: 0]  
Sequence number: 1734683627  
Acknowledgment number: 0  
Header Length: 40 bytes  
... 0000 0000 0010 = Flags: 0x002 (SYN)  
window size value: 5840  
[Calculated window size: 5840]  
Checksum: 0x41b4 [validation disabled]  
Urgent pointer: 0
```

# SNAT vs DNAT



# DNAT: DMZ



# DNAT: Port Forwarding

The screenshot shows the ASUS RT-AX88U web interface. The top navigation bar includes the ASUS logo, model name, power status, and links for Logout and Reboot. The main content area is titled 'WAN - Virtual Server / Port Forwarding'. It contains a description of port forwarding, a list of warnings, and a table of configured port forwarding rules. The 'Basic Config' section shows 'Enable Port Forwarding' is turned ON. The 'Port Forwarding List' table has one entry for 'mstsc' on port 111.

ASUS RT-AX88U

Powered by Asuswrt-Merlin

Logout Reboot English

Operation Mode: **Wireless router** Firmware Version: **384.19**

SSID: **Slysim\_jak mat...** **Slysim\_jak mat...**

Internet Connection Dual WAN Port Trigger Virtual Server / Port Forwarding DMZ DDNS NAT Passthrough

### WAN - Virtual Server / Port Forwarding

Virtual Server / Port forwarding allows remote computers to connect to a specific computer or service within a private local area network (LAN). For a faster connection, some P2P applications (such as BitTorrent), may also require that you set the port forwarding setting. Please refer to the P2P application's user manual for details. You can open the multiple port or a range of ports in router and redirect data through those ports to a single client on your network.

If you want to specify a Port Range for clients on the same network, enter the Service Name, the Port Range (e.g. 10200:10300), the LAN IP address, and leave the Local Port empty.

- When your network's firewall is disabled and you set 80 as the HTTP server's port range for your WAN setup, then your http server/web server would be in conflict with RT-AX88U's web user interface.
- When you set 20:21 as your FTP server's port range for your WAN setup, then your FTP server would be in conflict with RT-AX88U's native FTP server.

[Virtual Server / Port Forwarding FAQ](#)

#### Basic Config

Enable Port Forwarding ☒

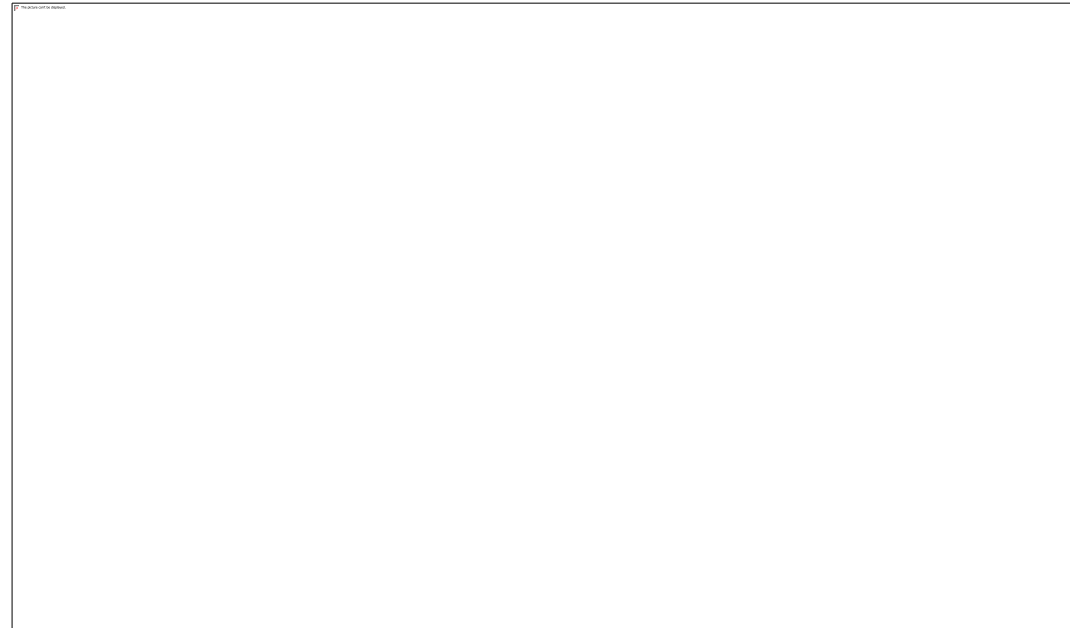
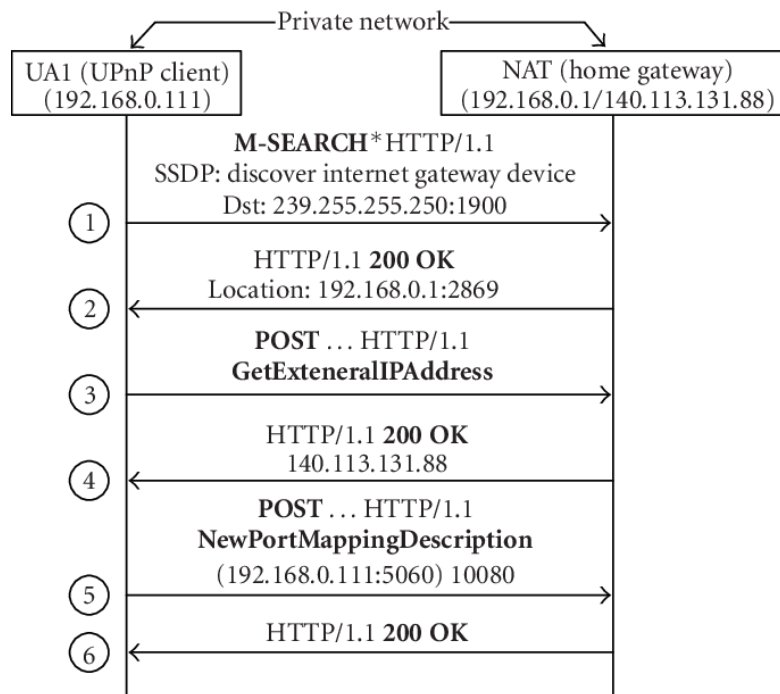
#### Port Forwarding List (Max Limit : 64)

| Service Name | External Port | Internal Port | Internal IP Address | Protocol | Source IP | Edit | Delete |
|--------------|---------------|---------------|---------------------|----------|-----------|------|--------|
| mstsc        | 111           | 3389          | 10.102.1.2          | TCP      |           |      |        |

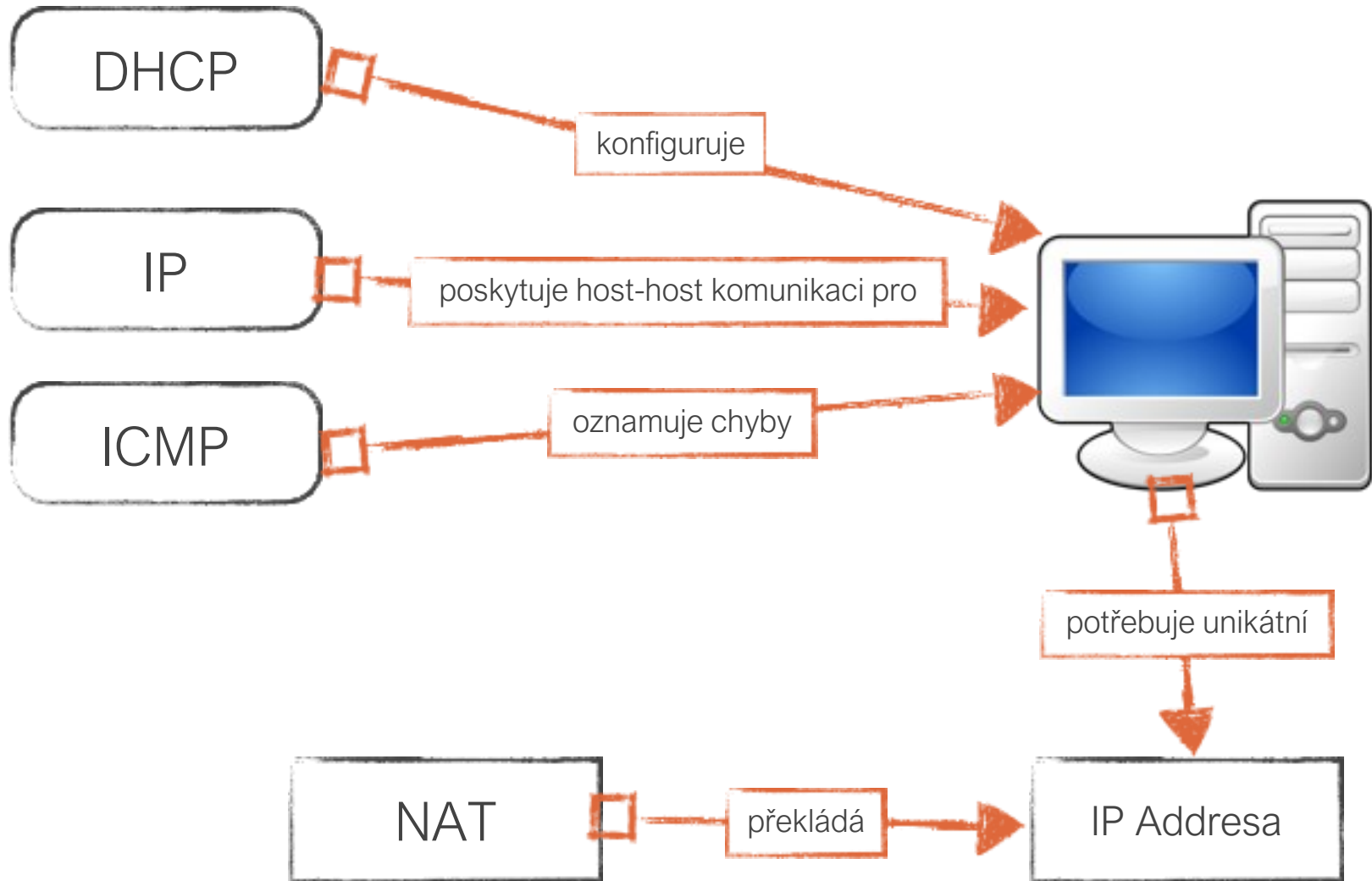
[Add profile](#)

# Překonávání NATu

- Dynamický Universal Plug and Play (UPnP)
- Internet Gateway Device (IGD) Protocol – TCP; STUN – UDP
- TURN



# Technologie



# Studijní materiály

- Kurose J.F., Ross K.W.: [Computer Networking, A Top-Down Approach Featuring the Internet](#). Addison-Wesley, 2003.
- Stevens, W.R.: [TCP/IP Illustrated, Volume 1](#). Addison-Wesley, 1994.
- RFC 791, 792, 950, 1918,